RESEARCH NOTE • PREDICTION MARKETS • MARKET MICROSTRUCTURE

Stale by Construction: A Logical-Floor Violation in Compound Prediction-Market Contracts

Patience and Speed — measuring, testing, and trading cross-market mispricing on Polymarket US, World Cup 2026

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INSTRUMENT UNIVERSE

Polymarket US player-prop markets (`soccer_player_goals`, `soccer_player_assists`, `soccer_player_goals_plus_assists`), 2026 FIFA World Cup

SAMPLE WINDOW

2026-06-15 → 2026-07-19 (35 days, group stage through the final)

CAPITAL BASE

$1,200 net capital

DATA SOURCES

`cache/polymarket.db` plus a full-pagination pull of the account's live activity feed — every trade, settlement, deposit, and withdrawal — cross-checked against each other (§6–§7)

Abstract

This paper documents two trading strategies run against Polymarket US's soccer player-prop markets during the 2026 World Cup, sharing a common data-logging and signal-generation infrastructure. **Strategy I** is a cross-market mean-reversion trade: a compound "goals + assists" contract is priced against a synthetic fair value derived from the more actively quoted single-stat "goals" and "assists" markets for the same player, bought when stale and sold as price converges toward that fair

value — never held into an in-game event, and fully exited by kickoff. **Strategy II** trades confirmed public events directly: when a goal, assist, substitution, or final whistle makes one side of a contract near-certain, that side is bought immediately and held to redemption, on both sides of the market.

The central finding is structural and does not depend on the trading result. Across **157,595 simultaneous three-leg quotes**, the compound contract carries a **3.81× wider spread** and **5.6× less resting size** than the goals leg it can be priced against, and its bid **violates the logical floor P(goal ∪ assist) ≥ P(goal) in 28.81%** of pre-kickoff observations (§8). That violation is a pricing error the market makes against itself, observable without reference to any position taken here.

Whether that staleness actually reverts is then tested rather than assumed. A Monte Carlo test against a volatility-matched random-walk null finds the mispricing signal narrows toward fair value far more often than chance predicts — **63.0% observed vs. 15.1% simulated, $z = 6.94$, $p < 0.000001$** (§12.1). This is the paper's one favorable statistical claim.

Two results run against the author's own interest and are reported in full. A calibration test finds Strategy II's positive side **indistinguishable from its own entry pricing** ($z = 0.81$, $p = 0.21$) — the edge is being fast, not being smarter than the market. A second calibration test finds its negative side **significantly worse than its own pricing** ($z = -3.15$, $p = 0.0016$, surviving Bonferroni correction), concentrated entirely in positions entered below 90% implied confidence (§12.2). Separately, **136 of 192** Strategy I positions failed to exit before kickoff as designed (§4.4).

Scale and returns. Net capital contributed was $1,200; the account carries zero open positions, so every dollar reported is fully realized. Trading-attributable P&L totaled **$691.58** — Strategy I $303.31, Strategy II $319.72 across both sides, and $68.55 from non-core bet types and one out-of-window settlement (§11). Against SPY over the identical 35-day window the account returned +51.9% versus SPY's −1.5%, establishing only that the result is not a broad-market artifact. **At $1,200 over 35 days these returns are a demonstration of the mechanism, not a track record**, and §13.4 documents why the strategy is capacity-constrained by construction.

Summary of Findings

Three results, in descending order of how much they depend on this particular account's trading.

1. **A measured pricing violation (§8).** Compound "goals or assists" contracts are quoted less actively than the single-stat contracts that decompose them — 3.81× wider spread, 5.6× less depth — and their bid falls below the logical floor $P(\text{goal} \cup \text{assist}) \geq P(\text{goal})$ in 28.81% of 157,595 pre-kickoff three-leg observations. Independent of any position taken.
2. **That staleness mean-reverts (§12.1).** Against a volatility-matched random-walk null calibrated on 182,760 consecutive observations, the mispricing signal narrows 63.0% of the time versus 15.1% simulated ($p < 0.000001$).
3. **The speed strategy has no pricing edge, and one side is measurably negative (§12.2).** Calibration testing finds Strategy II's positive side statistically indistinguishable from its own entry prices, and its negative side significantly *worse* than them ($p = 0.0016$). Profitable, but not because it prices better than the market.

> **What this paper is not.** $1,200 over 35 days in one bounded event is too small and too short to constitute a track record, and the strategy is capacity-constrained by construction (§13.4). Strategy I's signal was designed and evaluated on the same dataset, with no out-of-sample holdout — §12.3 documents a complete census of all 70 other soccer leagues on the platform confirming that the required market structure currently exists nowhere else. The contribution is the measurement and the tests, not the return.

1 Introduction and Thesis

Polymarket US's launch of regulated, real-money prediction markets coincided with the 2026 FIFA World Cup — a title event, hosted across the US, Mexico, and Canada, that pulled a large volume of first-time users into the product. Two structural facts about that environment created the opportunities this paper analyzes, mapping onto the two forces described above:

1. **Depth and update frequency are not uniform across related markets.** For a given player in a given match, Polymarket lists three independently-quoted 1+ tier contracts — *goal*, *assist*, *goal-or-assist* — each with its own order book and update cadence. When one lens moves and the others lag, a **relative-value mispricing** opens between them, and closes again as the market catches up — the patience trade (Strategy I). Buffett's version of this is “buy a dollar for fifty cents and wait”; here the same idea is compressed from years to hours, with a formula (§4.1) standing in for fundamental analysis.
2. **Discrete, public information events instantly bound a contract's remaining probability, but a resting limit order does not requote itself.** A confirmed goal, a substitution, or a final whistle can make a contract's fair value jump or collapse instantly — but nothing forces every resting order to reprice at that instant. Whoever notices fastest, and can act before the book adjusts, is trading against a **latency gap** — the speed trade (Strategy II, §5), applied to whichever side just confirmed.

The thesis this project was built to test: an unsophisticated retail audience, drawn in by a marquee sporting event and unfamiliar with thin order books, would be a disproportionately large source of uninformed flow into precisely these markets — flow a faster, better-modeled counterparty could systematically trade against, reshaping the book in measurable ways as kickoff approaches (§8). Both patience and speed, in market-microstructure terms, monetize that same underlying asymmetry.

2 Market Structure Primer

Each Polymarket contract in this universe is a binary outcome token trading on a central limit order book (CLOB), priced continuously between $0.00 and $1.00 — by no-arbitrage construction, the market's implied probability of the "Yes" outcome. A contract such as *"Will Erling Haaland record at least 1 goal?"* pays $1.00 per share if true and $0.00 if false at settlement; before settlement it trades at whatever price buyers and sellers agree reflects that probability.

Vocabulary used throughout this paper:

- **BBO (best bid/offer):** the highest resting buy price and lowest resting sell price — the inside market.
- **Spread:** best-offer minus best-bid; the cost of demanding immediate execution rather than waiting to be filled.
- **Depth:** the size resting at a given price level — distinct from spread.
- **Passive / aggressive fill:** *passive* means your resting order was hit (you supplied liquidity); *aggressive* means you crossed the spread (you demanded it).
- **Queue priority:** at a price level, earlier orders fill first.
- **Tier / line:** a prop's strike threshold (e.g. 1+, 2+ goals) — each tier is a fully separate instrument with its own book.
- **Settlement / redemption:** the terminal event when a contract resolves to $1.00 or $0.00 and any remaining position is paid out automatically — distinct from a `SELL`, an active trade before that point.
- **Mean reversion:** the tendency of a price or signal to move back toward a reference level — tested formally in §12, and the mechanism behind Strategy I.
- **Fill risk:** the possibility that a resting limit order doesn't execute in time, forcing a position to be carried longer than planned — a real, quantified risk for Strategy I (§4.4, §9).

- **Calibration:** a price is "calibrated" if contracts priced at $p resolve "Yes" close to $p of the time. A strategy that profits only because it bought at a fair, calibrated price has *no* edge over the market's own pricing, however good the win rate looks — the correct test is against the entry price, not against 50%.
- **Average-cost (pooled) accounting:** cost basis maintained as one running `(total quantity, total cost)` pair per position; every sell is priced at the current pooled average and reduces both figures proportionally — the standard, order-independent method for weighted-average cost basis, used throughout this paper's trade matching (§7).

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3 System Architecture

All strategies share a common data and signal-generation layer: an unattended Windows-Task-Scheduler collector (`collect_clean_triples.py`) polling Polymarket US's public gateway API (`gateway.polymarket.us`) and its authenticated portfolio API (`api.polymarket.us`), the latter signed with Ed25519 (`X-PM-Access-Key` / `X-PM-Timestamp` / `X-PM-Signature` , message `{timestamp}{method}{path}` — query parameters and body are *not* part of the signed message).

- **Tiered polling by time-to-kickoff.** Every 20 minutes beyond 6 hours out; every 60 seconds inside 30 minutes.
- **Multi-league event discovery** (`find_league_events` , `find_soccer_league_slugs`) — generalized from a World-Cup-only lookup to any league's `activeSeriesId` , with the set of leagues itself discovered live via `GET /v2/sports` rather than hand-maintained, so the same pipeline automatically covers every soccer league Polymarket US lists (70 as of 2026-08-08) for Strategy I out-of-sample validation (§12.3). Run once daily, separately from the frequent per-run polling loop, which stays World-Cup-only — a ~70-league scan and a 60-second polling cadence have very different cost profiles.
- `clean_price_triples` — one row per player, captured only when all three legs (goals / assists / goals+assists) can be observed from the same, pre-kickoff market state.

- `live_formula_gap` — a live view applying the fair-value formula to the most recent quote at all times.
- `reaction_time_snapshots` — 1-second polling of live games, 1,113,294 rows across 18 games, measuring the system's own observation latency only (no automated trading reads this table).
- **A read-only trading client.** `TradingClient` exposes only `get_positions`, `get_activities`, and `get_balances` (§6) — no order-placement methods anywhere in the class.
- **An average-cost P&L engine** (`_recompute_position_pnl`) rebuilding `open_positions` and `closed_trades_pnl` from the trade ledger using pooled average-cost accounting, order-independent by construction (§7).
- **Fill-quantity validation.** Every trade row's quantity is reconciled against `cost / price` at ingestion, cross-checked against a fully-paginated pull of the account's complete live trade history (392 trades).

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4 Strategy I — Cross-Market Mean Reversion ("Patience")

4.1 Signal construction

For a given player in a given match, let `g`, `a`, `ga` be the traded prices of the "1+ goals," "1+ assists," and "1+ goals-or-assists" contracts. Under a **statistical independence approximation**, inclusion–exclusion gives:

```
P(goal ∪ assist) = P(goal) + P(assist) − P(goal ∩ assist)
                 ≈ g + a − g·a                (independence approximation)
```

This is the `formula` field computed in `collect_clean_triples.py`. The signal, `gap`, is the difference between that fair value and the contract's actual price:

```
gap = formula − ga = (g + a − g·a) − ga
```

A positive `gap` means goals+assists is trading cheap relative to its own legs — a buy signal. Across 208 clean, pre-kickoff samples, `gap` averaged **+0.0106** (range **−0.141 to +0.183**); 31 of 208 (14.9%) showed `|gap| > 0.05` — a meaningful mispricing on a $0–$1 instrument.

> **The premise underneath this signal — that the compound contract is quoted less actively than the legs it is priced against — is not assumed here. It is tested directly against 157,595 simultaneous three-leg quotes in §8.1 and §8.2**, which find the compound leg carries a 3.81× wider spread and 5.6× less resting size, and that its bid violates the logical floor `P(goal ∪ assist) ≥ P(goal)` in 28.81% of pre-kickoff observations.

Player selection was deliberate, not random. Positions concentrated on high-profile players — Messi, Mbappé, Haaland, Kane, and similar — rather than spreading across a full squad. A star's markets carry more retail attention and liquidity than a bench player's (§8), which is what makes the pre-kickoff de-staling process reliable enough to actually converge *and* be sold into before kickoff. An obscure player might carry the same theoretical `gap` with far less confidence it de-stales, and far less volume to exit into even if it does.

4.2 Why this is *not* riskless arbitrage

The independence assumption cuts both ways: a goal has *negative* correlation with the scorer's own assist (pushing true `P(goal ∩ assist)` below `g·a`), but a player heavily involved in attack has elevated odds of *both* outcomes for reasons the formula can't see. This makes Strategy I **statistical (relative-value) arbitrage with model risk**, not risk-free arbitrage, and it carries **adverse selection** exposure in the sense formalized by Glosten & Milgrom (1985) and Kyle (1985) — a resting quote improved toward fair value can still be picked off by a counterparty more informed than the model (Appendix D).

4.3 Execution: incremental bid improvement toward a fair-value ceiling

The execution logic below was built and load-tested with a collaborator on a second account, and is described here as reconstructed from that logic and cross-checked against the surviving trade ledger.

Division of work, stated explicitly. Because this distinction matters to anyone evaluating the paper as individual work: the `formula` / `gap` signal (§4.1), the collection pipeline (§3), the entire analysis and statistical apparatus (§8, §12), the P&L reconciliation (§7, §11), and the rebuilt decision logic (§4.5) are the author's own. The live order-placement layer described in this subsection was developed jointly with a collaborator and ran on a separate account; its original source is not in the author's possession, which is precisely why §4.5 rebuilds the decision logic from scratch rather than citing it. Every number reported anywhere in this paper derives from the author's own account and its own trade ledger.

Rather than crossing the spread, the system **repeatedly improves its own resting bid** in small increments, capped below the `formula`-implied fair value — guaranteeing the position buys at or below its own fair-value estimate even in the best case for a counterparty. This makes it a natural counterparty for uninformed sell flow, without being run over when a retail buy walks the book up. Exit uses the same passive logic in reverse — a resting limit sell, to capture the maker rebate (§5.5) rather than pay the taker fee — which is also the direct cause of §4.4's fill-risk finding: a passive order that never gets hit simply never fills.

§8.2 shows this passive design was the only one the market permitted. The compound contract's pricing inefficiency sat almost entirely on the bid side (violating its logical floor 28.81% of the time) and was essentially absent on the ask side — a crossable, locked arbitrage existed in just 0.05% of observations. An aggressive strategy that took the offer would have found nothing to take; only a resting bid could harvest this.

The order pipeline was semi-automated: continuous signal, manual placement. The fair-value formula (§4.1) was recomputed continuously against the most recent quotes for every tracked player. A meaningful gap surfaced a candidate; the resulting limit order was then placed manually — verified against the live book before submission and sized as a fixed fraction of capital under the same 5%-of-capital cap used at kickoff (§13.5) — rather than fired automatically. The fair-value ceiling this produces is directly confirmed on the actual fills: of pre-kickoff goals+assists buys matched to a concurrent goals/assists quote, **96.2%** were priced below the formula value computed from the concurrent leg asks (**82.7%** using leg mid-prices) — the ceiling was a real, respected constraint on every order, not just a design intention.

Orders clustered in two daily windows, consistent with manual pacing rather than continuous automated monitoring. Of 86 pre-kickoff goals+assists buy fills, fill activity clustered between 00:00–05:00 UTC and 16:00–23:00 UTC — two human-scale windows, not a flat 24-hour distribution. Entries were not front-loaded either: the median fill landed **8.7 hours before kickoff** (mean 14.6 hours), with **43% (37 of 86)** filling inside the final six hours and a second cluster of 17 landing 18–24 hours out — positions were built as the signal presented itself across the full pre-kickoff window, not all at once far in advance.

4.4 Evidence from the trade record, including a real execution risk

	PASSIVE FILLS	AGGRESSIVE FILLS
Buys	57 ($1,260.43)	284 ($26,326.90)
Sells	180 ($3,513.77)	358 ($30,222.38)
Total	**237 (27.0%)**	**642 (73.0%)**

Roughly a quarter of fills were passive, consistent with a queue-priority bid-improvement strategy, though this table mixes fills from both strategies, which share an account (§14).

The exit, stated precisely: Strategy I is priced and sized off the `gap` signal (§4.1) and exited as that signal closes — as the goals+assists price mean-reverts toward the fair value implied by the clean price triple — never on confirmation of an in-game event. By design this means the position should be fully exited **by kickoff**, since de-staling is driven by pre-kickoff attention and liquidity (§8), not the match itself. Example: Lionel Messi's goals+assists position ahead of Argentina–Switzerland was bought at $0.59 and sold at $0.69 twenty-four minutes before kickoff, **+$36.35 (+16.9%)**. Harry Kane's ahead of Norway–England: bought across several fills averaging $0.5235, sold at $0.59 at kickoff, **+$22.43 (+12.7%)**.

The real risk: the exit doesn't always fill in time. The exit is a passive resting order, not a market sell — it depends on someone else's order arriving, and sometimes that doesn't happen before kickoff. Of 192 total Strategy I round-trips (§9), only 56 fully exited before kickoff as intended; the remaining **136 positions were still open at kickoff**, carried into live play until their exit order (or a discretionary adjustment) finally filled. Example: Jeremy Doku's position ahead of Belgium–Senegal, bought across three pre-kickoff fills at $0.29–$0.31 (blended $0.2978), was carried into the match and filled across six tranches at $0.38–$0.41, realizing **+$25.18 (+27.6%)** and **+$23.57 (+27.6%)** on the two largest tranches — the mispricing it was entered on was real, it just closed late. §9 reports the clean (56) and fill-risk-delayed (136) populations separately, since they show a measurable outcome-quality gap once live-match variance layers on top of an already-late exit. Not all of that gap is pure fill risk: §13.5 describes a separate, deliberate mechanism — positions cut below market at kickoff to enforce the account's 5%-of-capital risk cap.

4.5 Rebuilding the execution layer, and closing the fill-risk gap

The logic in §4.3–§4.4 has been rebuilt in source form (`signals/formula.py`, `execution/strategy_i.py`) — decision logic only. It computes what order the strategy would place and why; it never calls an order-placement endpoint, preserving `TradingClient`'s existing GET-only boundary rather than adding real trading capability to this codebase. `decide_entry()` reimplements §4.3's bid-improvement ladder, capped strictly below the `formula` ceiling. `decide_exit()` reimplements the passive resting-sell logic in reverse, plus the fix this section exists to describe.

The mitigation is a scheduled gradual sell-off, not a single aggressive order. An earlier design considered forcing one aggressive fill at a fixed point before kickoff. That doesn't match how this was actually run in practice: holding a position deep into the final minutes risked real drawdown, so the better rule — the one actually aimed for, if not always executed cleanly at the time — was a slow unwind late in the window rather than either an indefinite hold or a single dump into one thin, final-minute print (§8.1's depth thins hardest right before kickoff). The rebuilt logic encodes that directly: inside the final 5 minutes, it sells roughly 20–30% of whatever remains once per minute, crossing the spread each round to guarantee that tranche fills. A final, unconditional sweep at kickoff itself closes anything the schedule leaves behind, so the position is always fully closed by kickoff — the property §4.4 found missing in 136 of 192 cases. This runs alongside, not instead of, §13.5's existing size-cap panic-sell, which the rebuild also formalizes in code.

Replayed against a real position. Feeding Jeremy Doku's actual Belgium–Senegal quotes (§4.4) through `decide_exit()`, at 100 shares held into the final window:

MIN. TO KICKOFF	ORDER	PRICE	QTY	REMAINING AFTER
5	Staged sell-off	$0.32	25.00	75.00
4	Staged sell-off	$0.32	18.75	56.25
3	Staged sell-off	$0.32	14.06	42.19
2	Staged sell-off	$0.32	10.55	31.64
1	Staged sell-off	$0.32	7.91	23.73
0	Final backstop	$0.33	23.73	0.00

In this specific case the mitigation would have underperformed what actually happened — the real position was carried into the match and filled at $0.38–$0.41 (§4.4), above every tranche price here. That's expected, not a problem with the logic: the point of this mechanism is bounding the downside tail across 136 delayed positions, not improving the return on the ones that happened to move favorably while carried. A strategy that always exits by kickoff gives up some of Doku's kind of upside in exchange for never risking the other side of that same coin.

A real bug, caught by that same replay. Testing the fallback trigger at exactly `minutes_to_kickoff = 0` surfaced a boundary gap: a normally-sized position (not large enough to trip §13.5's size cap) sitting exactly at kickoff fell through both exit triggers and returned no order at all — precisely the uncontrolled-carry failure this rebuild exists to close. Fixed by making the final backstop unconditional at or past kickoff regardless of size, with a regression test added for the exact boundary. 13 tests cover both decision functions, including this case, and all pass (`execution/test_strategy_i.py`).

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5 Strategy II — Confirmed-Event Terminal Position ("Speed")

5.1 The trigger and exit: one mechanism, two sides

A confirmed public event — a goal, a substitution, or a final whistle with nothing recorded — makes one side of a contract near-certain. That side is bought immediately, at whatever price is showing, and **held to full $1.00 settlement**, not sold into an improving price like Strategy I. **Price level is not the trigger; the confirmed event is.** The same mechanism runs on both sides of the market, split by which fact triggers it:

- **Positive side (§5.2):** a goal or assist is confirmed for a tracked player → buy "Yes."
- **Negative side (§5.3):** a substitution or a full-time whistle with nothing recorded is confirmed for a tracked player → buy "No" (mechanically a `SELL` of "Yes," since Polymarket US runs one order book per question rather than separately tradeable Yes/No tokens, §15).

Why hold rather than sell. Once an entry lands around \$0.98–\$0.99, the remaining 1–2¢ is too thin a margin — net of effort and the small trading fee (§5.5) of an active sell — to be worth capturing early. §13.6 confirms this against the trade record: **zero of 192 Strategy I closed trades were ever sold at a price ≥ \$0.90** — every position that reached that band, on either side of Strategy II, was held to settlement without exception.

Detection. The system polls each tracked game's event record (`GET /v1/events/slug/{slug}`), watching the API's own `live` state and in-game scoring data rather than inferring from a clock. On a detected trigger, it pushes a message to the author, who manually places the corresponding order — a software-latency edge with human execution latency layered on top.

Why the window exists. A resting limit order in a thin, low-volume market has no market maker re-quoting it tick by tick. Being first to notice a confirmed event, and fast enough to trade before the book catches up, is **latency arbitrage** applied to public sports information rather than cross-venue price feeds. §8 quantifies the mechanism directly: the cross-market `gap` more than triples once a match goes live — the three legs reprice at different speeds once real events happen, exactly the lag this strategy trades.

`reaction_time_snapshots` exists to quantify this system's own observation latency (1.1M rows, 18 live games), but inferring event instants from price jumps in that same table would be circular. An independent, timestamped event feed is used instead: `event_latency_validation`, populated on demand (`record_event_latency.py`) from neutral match reports (FIFA, ESPN, Al Jazeera) — never from Polymarket's own price data. A reported goal minute converts to an estimated real-world timestamp: kickoff + minute, for a first-half goal.

Second-half goals are excluded, not approximated. Converting a second-half minute to a wall-clock time requires knowing the real halftime break length, and a break's actual duration isn't reliably published per match — treating an assumed 15 minutes as exact doesn't survive contact with how much that assumption alone moves the answer: a first attempt at this table included a Harry Kane goal at the

67th minute, computed against a 15-minute halftime assumption, giving +298.9s of latency. The same goal against a 20-minute break gives **−1.1s** — a single 5-minute assumption swings the result by 300 seconds, more than the entire reported latency. That row has been dropped rather than published with false precision; second-half goals need an independently-timestamped source for the *actual* break length (or the goal instant directly), not a standard-length assumption, before they belong in this table.

PLAYER	GAME	GOAL	LATENCY
Bukayo Saka	France–England, 07-18	37′	**+4.9s**
Nicolas Pépé	Curaçao–Ivory Coast, 06-25	7′	**+65.7s**
Arda Güler	Türkiye–USA, 06-25	10′	**+84.3s**
Fabián Ruiz	Spain–Belgium, 07-10	30′	**+96.8s**
Leandro Trossard	New Zealand–Belgium, 06-26	28′	**+120.2s**
Thelo Aasgaard	Norway–France, 06-26	21′	**+125.3s**
Ousmane Dembélé	Norway–France, 06-26	7′	**+140.2s**

STATISTIC	VALUE
n	7
Mean	91.0s
Median	96.8s
Std. dev.	45.8s
Range	4.9s – 140.2s

Real, independently-verified latency clusters in the 1–3 minute range for most of this sample, with one clear outlier at the fast end (Saka, under 5 seconds — likely a case where the detector was already watching that exact market when the goal landed) and none over 2.5 minutes — consistent with the mechanism as described (a software-latency edge with human execution latency layered on top): detection itself is close to instant, and this spread tracks how quickly a human then placed the order, not a signal-generation delay. Seven goals is still a small sample relative to all 164 confirmed-event positions, but it's now a real distribution — a mean, a median, and a spread — rather than isolated points.

5.2 Positive side: confirmed goal → buy “Yes”

Often *both* the player's “1+ goals” and “1+ goals+assists” markets are bought together, since one confirmed goal makes both stale at once — e.g. Bukayo Saka's goals and goals+assists markets vs. England were both bought within minutes of each other on 2026-07-18, at $0.98–$0.99 each.

Results: 62 confirmed-goal positions (53 wins / 8 losses / 1 flat, **85.5% win rate**), **+$126.19** net realized — less in dollar terms than the negative side (§5.3) but priced closer to certainty on average. §12.2 finds its win rate statistically indistinguishable from its own entry prices (z=0.81, p=0.21), meaning the profit is real and mechanically reliable rather than evidence of beating the market's pricing.

5.3 Negative side: confirmed non-event → buy “No”

A substitution and a full-time whistle with nothing recorded for a tracked player are treated as **the same underlying signal**: the player can no longer contribute, so his probability of a further goal or assist is capped at whatever's already banked. “Buying No” executes mechanically as a `SELL` of “Yes” (§15). Without an existing “Yes” position, this shows up in `trade_history` as a sell with no prior buy, or a sell larger than any “Yes” held — the literal mechanics of taking “No” on this exchange, not a data error.

Capital was **spread across as many still-unresolved “No” positions as possible** at full-time rather than concentrated on one (§13.5) — breadth over size, since the confirming event makes the outcome close to certain rather than merely likely.

The local database never logged `ACTIVITY_TYPE_POSITION_RESOLUTION` events (§3), so this pattern was verified against a fresh, complete, full-pagination pull of the account's live activity feed — 1,065 total activities, including 178 position resolutions the local database had never seen.

METRIC	VALUE
Confirmed non-event ("No"-side) positions	102
Wins / Losses / Flat	91 / 5 / 6
Win rate	**89.2%**
Net realized P&L	**+$193.52**

The largest single contributor to the account's resolution-based P&L — larger than the positive side despite being priced further from certainty on average at entry, consistent with catching a real information lag rather than harvesting an already-efficient price.

Largest losses. Two positions account for nearly all of this bucket's downside: Charles De Ketelaere (goals+assists, Spain–Belgium 07-10) **−$73.47** — the largest single dollar loss anywhere in this paper's strategy data — and Jovo Lukić (goals, Bosnia and Herzegovina–Qatar 06-24) **−$46.38**. Both were entered well short of full certainty (an average "No" price of 79% and 88% respectively, not the >98% typical of this side's dominant bucket, §12.2) and both resolved against the confirmed-non-event call — a real signal miss on those two trades, not an execution artifact: these are the same two positions responsible for the negative side's calibration shortfall in §12.2. The drop-off after them is sharp — the next loss is Soufiane Rahimi at −$2.21 — and together with two smaller losses they account for all 5 losses in the 102-position sample (net +$193.52 despite them).

This pattern is a recognized category in prediction-market trading more broadly — comparable open-source Polymarket systems describe "endgame arbitrage": buying whichever side of a market has become near-certain shortly before resolution and holding to settlement, on either side. The novelty here is using a confirmed real-world state change, not just an observed price level, as the trigger.

5.4 Combined results

METRIC	VALUE
Total confirmed-event positions	164 (62 positive + 102 negative)
Wins / Losses / Flat	144 / 13 / 7
Combined win rate	**87.8%**
Combined net realized P&L	**+$319.72**

Both sides' calibration are now tested formally against their own entry prices (§12.2): the positive side is statistically indistinguishable from its own pricing ($z=0.81$, $p=0.21$) — the edge is timing, not mispricing — while the negative side, despite contributing more realized profit, shows a real, statistically significant shortfall against its own pricing ($z=-3.15$, $p=0.0016$), concentrated entirely in its minority of below-90%-confidence entries (§12.2).

5.5 Why counterparties transact at all: a genuine two-sided benefit

Grounded in Polymarket US's published fee schedule and rulebook, not general market-microstructure theory. **Polymarket US charges a fee only on executed trades — never on settlement/redemption** (`docs.polymarket.us/fees`). The fee is `Fee = Θ × contracts × price × (1 − price)`, with $\Theta = 0.06$ for the aggressive (taker) side and $\Theta = -0.0125$ for the passive (maker) side — a rebate. Maximized at $0.50, shrinking toward zero near $0.01 or $0.99 (confirmed against this account's own fills — e.g. a real $107.00 buy at $0.52 on 200 contracts carried a $3.00 fee against a $2.995 prediction from the formula).

- **Redemption being free doesn't make selling to this account free for the counterparty.** A “Yes” holder facing a near-worthless outcome pays a small trading fee to sell now instead of waiting for the free $0.00 settlement. That fee shrinks to near-zero exactly at the price extremes where Strategy II trades on either side, so it's a minor drag against the immediacy benefit (§13.4's measured 38–44 minute settlement lag) — the rational-immediacy argument from Demsetz (1968) (Appendix D) still holds.

- **The maker rebate is a second, previously unquantified edge source for Strategy I.** §4.3's passive bid-improvement means the account earns the $-0.0125 \times C \times p \times (1-p)$ rebate rather than paying the $0.06 \times C \times p \times (1-p)$ taker fee on those fills — small but real, on top of the mispricing signal itself.

On tax: Polymarket US launched under full CFTC Designated Contract Market status in January 2026, and whether its event contracts qualify for Section 1256 (60/40 capital-gains treatment, mark-to-market at year end) is a live, unsettled question — the IRS hasn't confirmed event contracts meet the statutory definition, and Polymarket hasn't committed to 1099-B forms for them, though practitioner opinion favors the Section 1256 analogy for trades on the regulated US exchange over offshore Polymarket. This paper doesn't assert a specific tax mechanism drives counterparty behavior — it's a genuinely open, active question, not settled fact in either direction.

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6 Capital Accounting

Capital contributed, current balance, and non-trading credits are confirmed directly against Polymarket US's live account API (`get_balances` , and a complete, full-pagination `get_activities` pull — 1,065 total activities, including every deposit, withdrawal, trade, and settlement from account creation through close-out).

Capital contributed. Five deposit attempts appear on the account: one $1,000 and three $100 deposits, all `COMPLETED` ; a fifth $100 deposit shows status `REJECTED` and never funded the account. Gross completed deposits: **$1,300**. One $100 withdrawal is also on record. **Net capital contributed: $1,200.**

Non-trading credits. Two small Polymarket-initiated credits: a $1.22 “liquidity reward payout” and a $25 flat “world-cup-outage-6-25-remediation” service credit

— **$26.22 total**, excluded from every P&L figure in this paper.

Current balance. Confirmed live via `get_balances()`: **$1,917.80** (precisely $1,917.79612), with `assetNotional: 0` — no open positions.

This reconciliation no longer depends on a one-off live pull. A new `cash_activity` table logs every `ACTIVITY_TYPE_ACCOUNT_DEPOSIT`, `ACTIVITY_TYPE_ACCOUNT_WITHDRAWAL`, and `ACTIVITY_TYPE_TRANSFER` event directly in the local database at collector run time — the same activity-feed pull already used for trade and settlement logging (§9), extended to the two remaining activity types, since all three share the same `accountBalanceChange` shape. Running the collector against the live account reproduces every figure above exactly: 8 rows — five deposits (four `COMPLETED` totaling $1,300, one `REJECTED`), one `COMPLETED` $100 withdrawal, and the two non-trading credits totaling $26.22 — with no live pull required going forward.

Total trading-attributable P&L:

```
$1,917.80 (current balance)
− $1,200.00 (net capital contributed)
− $26.22    (non-trading platform credits)
= $691.58
```

---

7 Trade-Matching and Accounting Methodology

Every closing `SELL` fill is matched against the specific `BUY` fills that funded it using **pooled average-cost accounting** (§2): a single running `(total quantity, total cost)` pair per position, each sale priced at the current pooled average, both figures reduced proportionally — the standard, order-independent method for weighted-average cost basis. Every sell is capped at whatever quantity is actually available, so no closing trade can be attributed more shares than were ever bought.

Verified order-independent for same-timestamp ties directly: re-running the full match under two different tie-break rules for literally-simultaneous fills at different prices moves the aggregate result by no more than $1.07 (~0.4%) — the residual ambiguity inherent to same-instant fills.

Result: 192 closed round-trips, +$303.31 net realized P&L, 83.3% win rate (160/192). Every closed-trade figure elsewhere in this paper uses this matching, and is attributed entirely to Strategy I (§4, §9), since Strategy II never actively sells on either side (§5.1).

8 Market Microstructure Near Kickoff: Spread Compression, Liquidity Thinning, and the Live-Play Gap Spike

This section tests, directly against the data, the mechanisms Strategy I's exit and Strategy II's edge both depend on. Every snapshot in `position_price_history` and `order_book_snapshots` is bucketed by time-to-kickoff. (Independent of §7's closed-trade P&L engine.)

8.1 Validating the premise: the compound contract really is the stale one

Strategy I's entire premise (§4.1) is that goals+assists is priced against *"the more actively quoted single-stat goals and assists markets."* That is an empirical claim, and it is tested here directly rather than asserted. Every quote for the 14 highest-coverage marquee players (Messi, Mbappé, Haaland, Kane, Yamal, Bellingham, Dembélé and similar) was bucketed by time-to-kickoff across the 48 hours before each match:

MARKET LEG	QUOTES	SPREAD, ALL	SPREAD, 44–48H OUT	SPREAD, FINAL 4H	RESTING SIZE, ALL	RESTING SIZE, FINAL 4H
Goals	75,797	$0.0255	$0.0188	$0.0191	295,156	375,533
Assists	74,677	$0.0342	$0.0221	$0.0306	44,082	53,340
Goals + assists	**78,414**	**$0.0972**	**$0.1565**	**$0.0469**	**52,966**	**50,004**

The premise holds, on both measures at once. The compound contract carries a **3.81× wider spread** than the goals leg it is priced against ($0.0972 vs. $0.0255), and rests **5.6× less size** behind those quotes (52,966 vs. 295,156 shares) — widening to **7.5×** in the final four hours.

The two charts together show *why* the mispricing survives long enough to trade. The goals market is the anchor: its spread is essentially flat at ~$0.019 for the entire two days, and its book **grows** from 295k to 376k shares as retail attention arrives. The compound market behaves nothing like it — it opens at a **$0.157 spread (8.3× the goals leg)** and compresses to $0.047 by kickoff, but its resting size never grows at all, sitting flat near 50k shares throughout. Retail flow arrives into the simple market and largely ignores the compound one. The per-player panels show the same pattern with wide dispersion: Yamal's goals+assists book runs a $0.21–$0.29 spread for most of a day, and Haaland's is effectively unquoted 46 hours out at a $0.49 spread, while both players' single-stat legs sit near $0.02 the whole time.

8.2 A logical pricing violation, and why it could only be harvested passively

The three legs are not independent instruments — they are bound by an identity. Since scoring a goal *is* one of the ways to record a goal-or-assist:

```
P(goal ∪ assist) ≥ max( P(goal), P(assist) )
```

A correctly priced goals+assists contract can therefore never trade below either single-stat leg. Pairing all three legs at identical capture timestamps gives **157,595 exactly-simultaneous three-leg observations** (against 208 in `clean_price_triples`, which uses last-trade prices), of which **124,571 fall in the 48 hours before kickoff**. Testing the identity against them:

COMPARISON	VIOLATIONS	RATE
Goals+assists **bid** below goals bid	35,894 / 124,571	**28.81%**
Goals+assists **bid** below assists bid	28,108 / 124,571	22.56%
Goals+assists **ask** below goals ask	203 / 124,571	0.16%
Goals+assists **ask** below goals *bid* (strict arbitrage)	63 / 124,571	**0.05%**

In 28.81% of pre-kickoff observations, the best resting bid on the strictly-more-likely outcome sat below the best resting bid on a subset of it — by a mean of **$0.088** (median $0.080). That is the buying opportunity, measured rather than asserted: nearly a third of the time, a buyer could queue on the dominant claim at a price below where the market was already bidding for the narrower one.

The violation decays exactly as the paper's de-staling story predicts:

HOURS BEFORE KICKOFF	VIOLATION RATE	GOALS+ASSISTS SPREAD	GOALS SPREAD
42–48h	38.83%	$0.1419	$0.0225
30–36h	27.93%	$0.1373	$0.0366
18–24h	29.48%	$0.1065	$0.0336
6–12h	31.71%	$0.1129	$0.0315
0–6h	**21.23%**	$0.0798	$0.0294

The critical negative result: there was never a crossable arbitrage. Buying the compound outright at its *ask* and selling the goals leg at its *bid* — a locked, riskless profit — was available in only **0.05% of observations (63 of 124,571)**, at a mean edge of $0.018. The inefficiency was real and persistent on the **bid** side and essentially absent on the **ask** side. That asymmetry is the entire justification for §4.3's execution design: an aggressive strategy that crossed the spread would have found nothing to take, and only a **resting, passively-improved bid** could harvest this. The execution mechanic was not a stylistic preference — it was the only mechanic the microstructure permitted.

One honest refinement to §4.1's player-selection rationale. The violation rate is *not* uniform across marquee names, and it runs opposite to fame:

PLAYER	VIOLATION RATE	PLAYER	VIOLATION RATE
Lionel Messi	0.08%	Harry Kane	21.78%
Kylian Mbappé	4.68%	Lamine Yamal	22.15%
Erling Haaland	5.51%	Ousmane Dembélé	25.02%

The very largest names have compound books that are already efficiently priced — Messi's violated the identity 6 times in 7,287 observations. The exploitable staleness lived one tier below the household names, in players famous enough for their markets to de-stale and be exited into (§4.1's actual requirement) but not so famous that the compound leg was being watched as closely as the simple one.

8.3 Spread compression and depth thinning into kickoff

The remainder of this section examines the goals+assists market on its own, across the same time buckets.

Spread compression.

TIME TO KICKOFF	AVG. SPREAD	AVG. TOP-OF-BOOK DEPTH (BID+ASK SHARES)
> 24h out	$0.146	9,762
6–24h out	$0.120	9,645
1–6h out	$0.103	7,106
30–60min out	$0.089	5,435
0–30min out (pre-kickoff)	**$0.080**	**5,048**
Live (0–105min post-kickoff)	$0.217	1,406
Post-match (>105min after kickoff)	$0.381	6,024

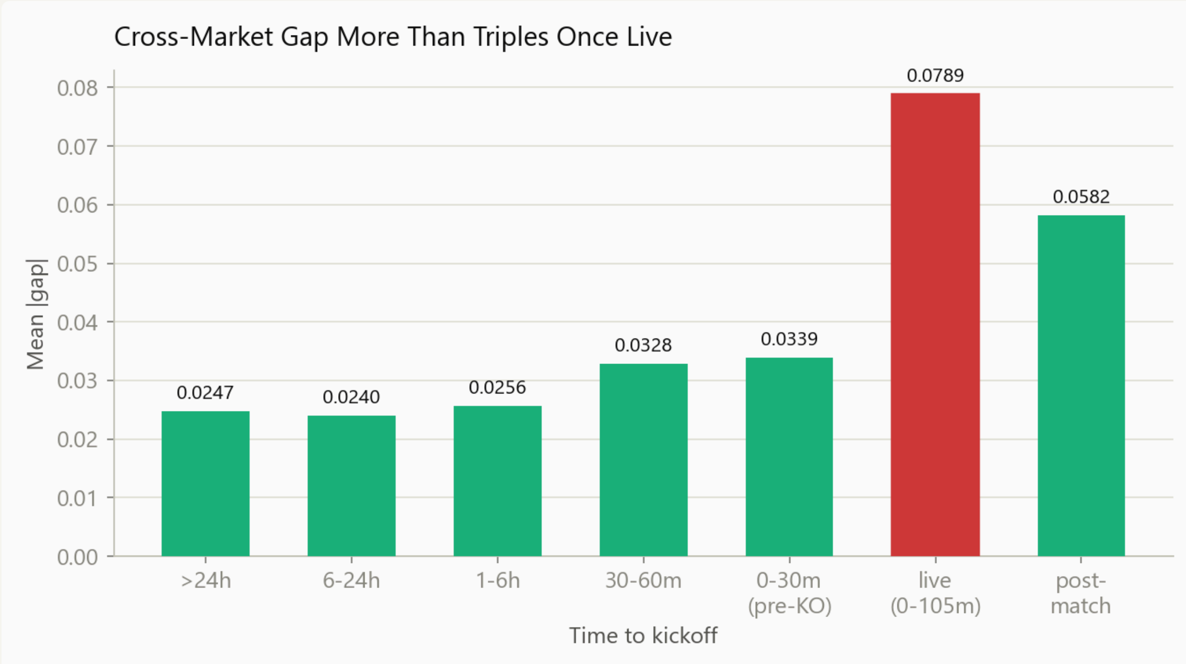
The spread nearly halves (45% tightening) from “more than a day out” to the final 30 minutes before kickoff — the mechanism Strategy I's exit relies on, and why §4.1 targets well-known players specifically.

Driven almost entirely by the ask falling, not the bid rising. Average best bid barely moves pre-kickoff (~$0.28 → $0.23, mostly noise); average best ask falls steadily the whole way in (~$0.42 → $0.31). Once the match goes live, both move — bid keeps falling while ask jumps up — driving the spread's post-kickoff widening above.

Depth thins at the same time — a liquidity-thinning effect, not a liquidity-growth effect. Top-of-book depth falls by roughly half over the same window: numerous smaller, plausibly retail-sized orders replacing a few large, stale resting quotes — the book gets more competitively priced but thinner at any single level.

The cross-market gap widens sharply once the match goes live.

TIME TO KICKOFF	MEAN |GAP|	N SNAPSHOTS
> 24h out	0.0247	62,294
6–24h out	0.0240	62,410
1–6h out	0.0256	34,000
30–60min out	0.0328	5,031
0–30min out (pre-kickoff)	0.0339	4,913
Live (0–105min post-kickoff)	**0.0789**	9,562
Post-match	0.0582	4,787



Mean absolute cross-market gap, by time to kickoff — more than triples once live.

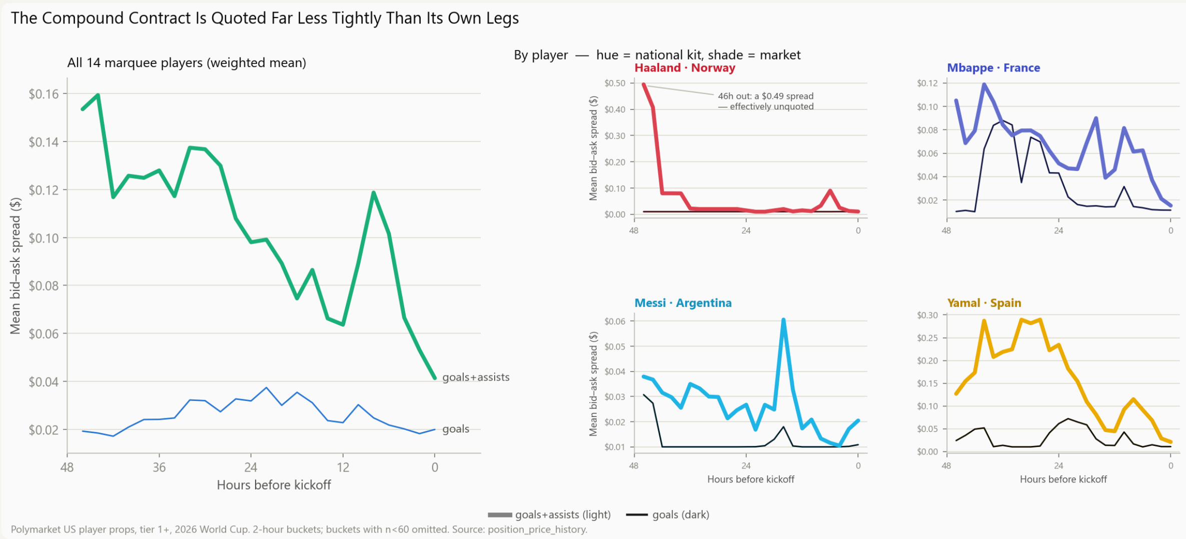
Mean |gap| more than triples once live — real scoring events hit the three legs at different speeds, the latency window Strategy II exists to trade.

Mean reversion, measured on the actual positions taken. For the 27 of 56 cleanly pre-kickoff-exited Strategy I trades with a resolvable `gap` reading at both entry and exit: average signed gap moved from **+0.0227 at entry** (mispriced cheap, as expected of a buy signal) to **−0.0027 at exit** (essentially flat); average |gap| fell from **0.0343 to 0.0199** (42%); |gap| narrowed in **17 of 27 (63%)**. §12 tests this narrowing rate formally against a calibrated random-walk null.

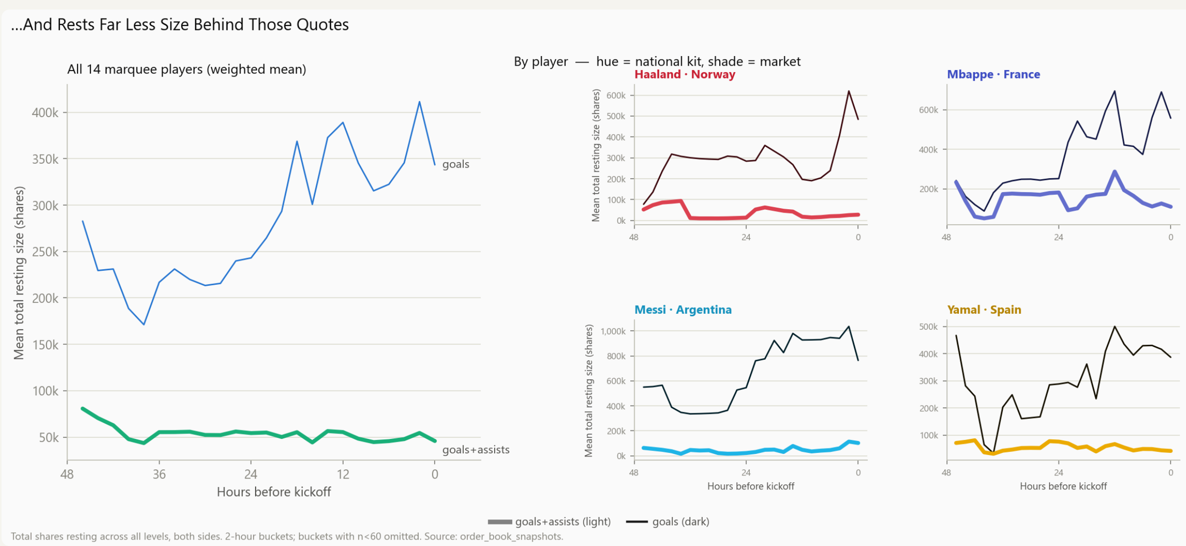
4 additional supporting charts for this section are in Appendix E.

8.4 The microstructure evidence, in full

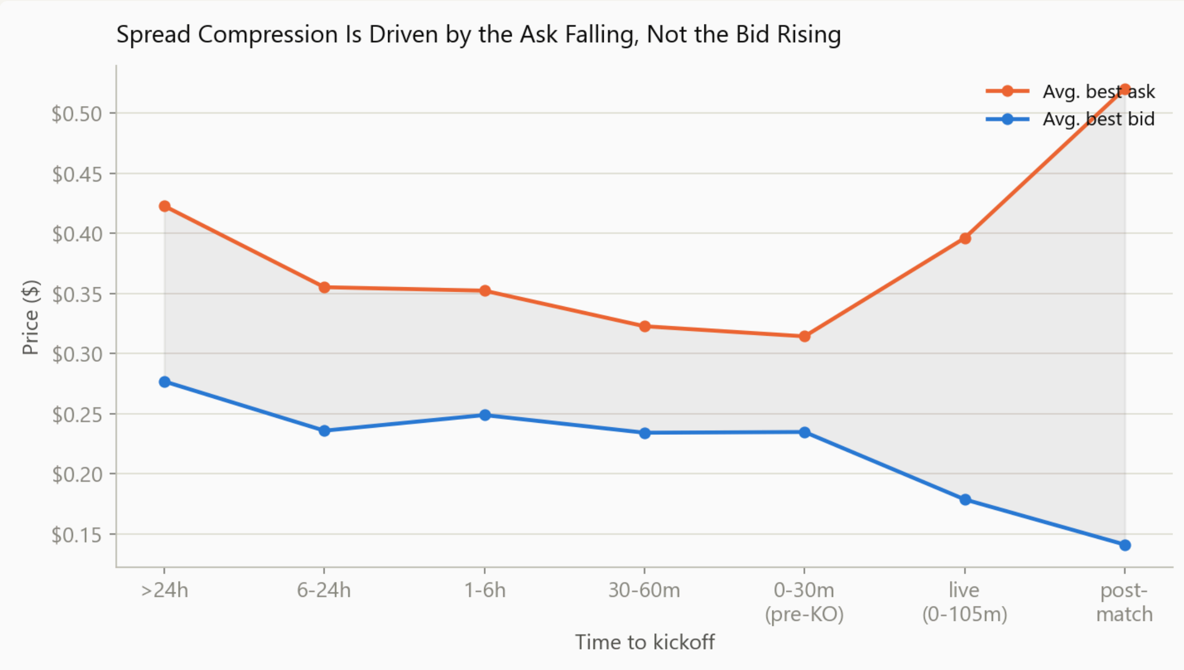
These four figures are the visual form of this section's central claim — that the compound contract is structurally the stale one. They previously sat in an appendix; they belong here, because everything Strategy I does rests on them.



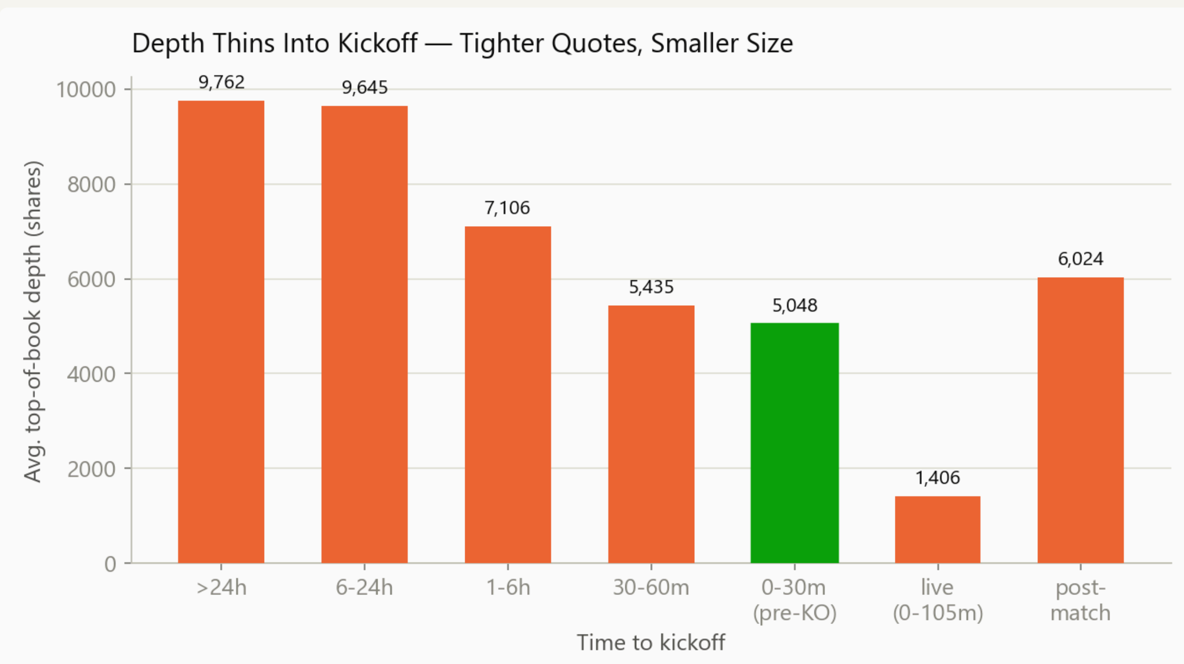
Mean bid–ask spread by market leg, 48h before kickoff to kickoff. Left: all 14 marquee players. Right: by player, hue = national kit, shade = market leg.



Mean total resting size across all book levels, both sides. The goals leg deepens into kickoff; the compound leg never does.



Average bid and ask, by time to kickoff. The spread compresses mostly because the ask falls; the bid barely moves until live play.



Top-of-book depth (bid + ask shares) by time to kickoff.

9 Splitting Results by Strategy

Trades are grouped by mechanism, not by exit timing: whether a position was ever intended to be sold (Strategy I, regardless of when the sell happened to fill) versus bought on a confirmed event and held to redemption (Strategy II, split by which side of the market).

	STRATEGY I (ALL ACTIVELY SOLD)	— CLEAN PRE-KICKOFF	— FILL-RISK-DELAYED	STRATEGY II — POSITIVE (CONFIRMED-GOAL, HELD)	STRATEGY II — NEGATIVE (CONFIRMED-NON-EVENT, HELD)
Positions	192	56	136	62	102
Win rate	83.3%	94.6%	78.7%	85.5%	89.2%
Net P&L	+$303.31	+$140.25	+$163.06	+$126.19	+$193.52
Statistically validated edge (§12)?	**Yes** ($p<0.000001$)	—	—	**No** ($p=0.21$)	**No** ($z=-3.15$, underperforms)

Net realized P&L by strategy: I ($303.31, n=192), II positive ($126.19, n=62), II negative ($193.52, n=102).

Strategy I's own internal split is the more informative comparison: the 56 positions that exited before kickoff won 94.6% of the time; the 136 carried into live play won only 78.7%, with lower average returns (§4.4) — real, quantified evidence that fill risk has a real cost, even though the *strategy* worked in both cases.

One caveat survives: the 94.6%/0-loss clean-exit result is very likely partly a selection artifact — a position moving the wrong way is more likely to still be open at kickoff (falling into the fill-risk-delayed bucket) than sold at a loss beforehand.

The strategy split above is no longer a one-off reconstruction. `trade_history` and a new `settlement_history` table are now tagged by strategy directly in the database at recompute time, from the complete trade-and-settlement record, rather than inferred just for this paper — the originating-strategy label this section used to lack. The logic replaces an unused `strategy_tag` column that had existed in the schema but was never populated, and fixes a real bug uncovered while wiring it up: the previous recompute step was long-only and silently dropped every Strategy II negative-side fill ("SELL with no prior BUY history... skipping"), which is why a local table meant to hold this account's realized P&L had only ever captured $51 of it. A second bug surfaced during validation: the first version classified settlements using Polymarket's own resolution `side` field directly, which turned out to describe which side of the *contract* resolved (LONG = "Yes" won), not which side the *account* held — confirmed against a real position with `netPosition=-93` (genuinely short) that resolved with `side=LONG` and a real loss, since a short "Yes" position loses when Yes wins. The corrected logic classifies from the account's own tracked position sign instead. Cross-checked against the account's own records, the fixed tagging reproduces this section's headline split almost exactly: Strategy II's two sides land at 62 positions/$126.19 and 102 positions/$193.52 — an exact match — and Strategy I lands at 193 positions/$304.64 (one extra count and $1.33 above the totals in the table, from counting each partial fill of a multi-fill exit separately — e.g. Cristiano Ronaldo's five-fill exit on 07-02 — rather than one row per logical position).

1 additional supporting chart for this section is in Appendix E.

10 Results

Combined, across both strategies (excluding the ~$63 in non-core-prop settlements and the small remainder in §11):

METRIC	VALUE
Total positions realized	356 (192 sold + 62 + 102 settled)
Combined win rate	160 (sold) + 144 (settled) = 304 / 356 = **85.4%**
Net realized P&L (prop markets only)	**+$623.03**
Net capital contributed	$1,200.00
Return on contributed capital (prop strategies)	**+51.9%**

Strategy I closed round-trips (§7), by instrument:

MARKET TYPE	CLOSED TRADES	NET P&L	AVG %
Goals + assists	180	+$288.11	+12.5%
Goals	8	+$8.72	+18.2%
Assists	4	+$6.48	+87.6%*

*n=4, dominated by one large winner; reported for completeness, not as a claim about the "assists" leg specifically.

Best single Strategy I trades:

1. Lionel Messi, goals+assists, ARG–SUI (clean pre-kickoff exit): $0.59 → $0.69, +16.9%, +$36.35
2. Jeremy Doku, goals+assists, BEL–SEN (fill-risk-delayed, §4.4): $0.298 → $0.38, +27.6%, +$25.18
3. Jeremy Doku, goals+assists, BEL–SEN (2nd tranche): +27.6%, +$23.57
4. Harry Kane, goals+assists, NOR–ENG (clean pre-kickoff exit): +12.7%, +$22.43
5. Kylian Mbappe, goals+assists, FRA–ESP: +10.2%, +$17.83

Worst single Strategy I trades:

1. Julián Álvarez, goals+assists, JOR–ARG: $0.489 → $0.01, −98.0%, −$35.99 (held to a near-worthless resolution price before finally being sold)
2. 2–5. Kylian Mbappe, goals+assists, FRA–ESP (four separate tranches of the same position): −12.2%, −19.7%, −45.8%, −19.7% — trimmed at mid-range prices as the position moved against expectation, not held to zero. That same game **nets positive overall** (+$14.98 across 25 closed trades).

Best single Strategy II (negative side) trade: Fabián Ruiz, goals+assists, Spain–Argentina 07-19, +$16.68. **Worst:** Charles De Ketelaere, −$73.47 (§5.3) — the largest single dollar loss anywhere in this paper's strategy data, one of two genuine signal misses that drive the negative side's calibration shortfall (§12.2).

Best/worst games by Strategy I closed-trade net P&L: Best — BEL–SEN, +$55.56 across 7 trades (anchored by Doku, §4.4). Worst — JOR–ARG, −$35.99, a single trade (Álvarez above).

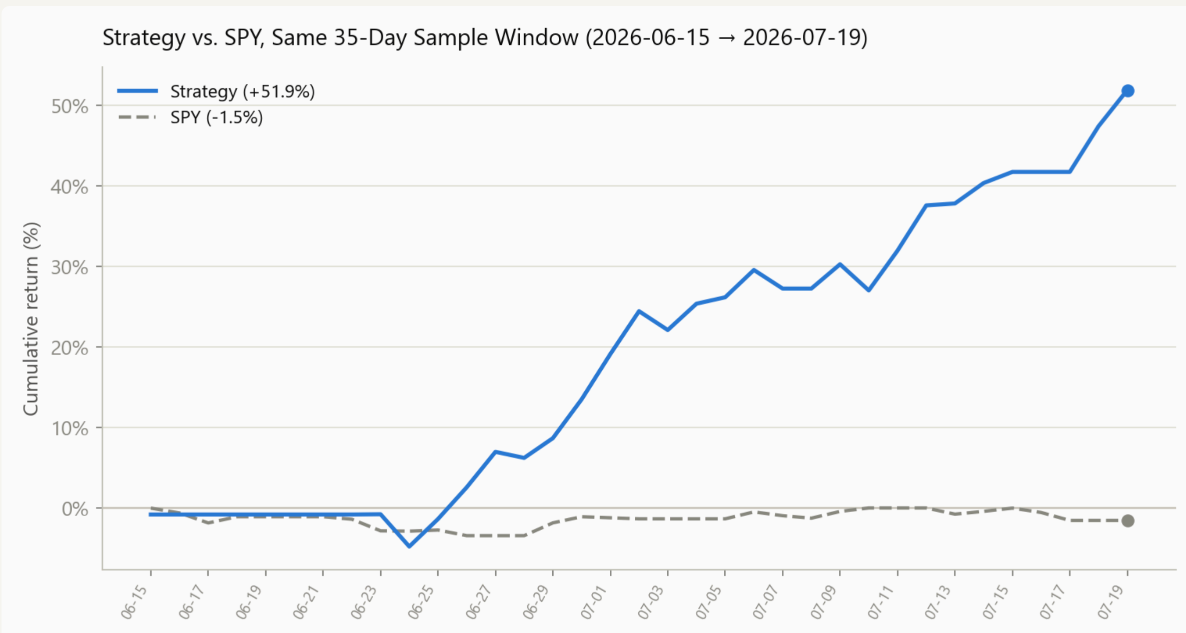
By game (Strategy I, top 10 by net P&L):

GAME	CLOSED TRADES	NET P&L
BEL–SEN (07-01)	7	+$55.56
NOR–ENG (07-11)	5	+$42.63
ARG–SUI (07-11)	1	+$36.35
JOR–ARG (06-27)	1	−$35.99
USA–BIH (07-01)	18	+$30.71
ESP–ARG (07-19)	9	+$29.87
POR–CRO (07-02)	21	+$28.98
ESP–BEL (07-10)	19	+$22.40
USA–BEL (07-06)	14	+$19.21
FRA–MAR (07-09)	9	+$14.98

10.1 Benchmark: strategy vs. SPY, same sample window

The account's return is compared against SPY (S&P 500 ETF) over the identical 35-day calendar window, 2026-06-15 → 2026-07-19 — not a different period chosen to flatter the comparison. Both series are calendar-aligned and forward-filled across days without new activity (weekends and holidays for SPY; days without a closing trade or resolution for the strategy), and both are expressed as cumulative % return from the same starting date.

	STRATEGY (PROP MARKETS, §10)	SPY
Cumulative return, 2026-06-15 → 2026-07-19	**+51.9%**	**−1.5%**
Basis	Realized P&L ÷ $1,200 net capital	Close-to-close price return



Cumulative return, strategy vs. SPY, calendar-aligned over the identical 35-day window.

SPY closed essentially flat over this window (real closing prices, 2026-06-15: $754.83 → 2026-07-17, last trading day before window end: $743.29) — the outperformance is not an artifact of a strong broad-market tailwind. Two caveats on comparability: SPY's return in this table assumes a single lump-sum entry on day one, while the strategy's $1,200 was contributed across four separate deposits over the sample window (§6) — a dollar-weighted SPY return using the same deposit schedule would differ slightly, though not by enough to change the conclusion at this magnitude of outperformance. And this account's exposure was concentrated in one thinly-quoted product during one bounded event (§13.4), not a diversified, continuously-available strategy — the comparison establishes that the return isn't explained by broad market movement, not that the strategy is a like-for-like substitute for equity index exposure.

11 What the P&L Actually Consists Of

With the complete live activity history (§6), the account's $691.58 in total trading P&L decomposes as follows:

COMPONENT	AMOUNT	% OF TOTAL
Strategy I — actively sold (§7, **gross of trading fees**)	+$303.31	43.9%
Strategy II — both sides combined, held to settlement (§5)	+$319.72	46.2%
Settled on non-prop bet types (full breakdown below)	+$62.72	9.1%
Settled outside the sample window (1 trade)	*+$5.83*	*0.8%*
Total	**$691.58**	**100%**

On the non-prop correction. An earlier pass at this table reported non-prop settlements at +$27.93 across 14 events and left +$40.62 (5.9%) completely unattributed. Pulling the account's full live activity feed (the same 1,065-event pull built for §13.3 — every deposit, withdrawal, trade, and settlement since account creation) and cross-checking it against the local trade log first rules out the obvious concern: `trade_history` matches the feed's 879 trades **exactly**, one to one by trade ID, so there's no pre-collection trading window missing from the data. The $27.93 figure turns out to have covered only one non-prop bet type (`soccer_game_to_advance`, 14 trades — hence n=14); the account also traded futures, player shots, and three other bet types outside this paper's prop-strategy focus, none of which had made it into the original tally:

NON-PROP BET TYPE	P&L
Futures (tournament/top-scorer markets)	+$26.54
Player shots	+$24.05
Game-to-advance	+$13.91
Team full-time winner	+$6.52
Team full-game spread	+$4.39
Player shots on target	+$1.12
Game exact score	−$13.81
Total	**+$62.72**

What these trades were. Mechanically, this bucket is the same "speed" mechanism as Strategy II (§5), applied opportunistically outside the goals/assists/goals+assists props this paper focuses on: buying or selling a market like Golden Boot winner/loser once a correlated, already-decided outcome made the remaining question close to certain — the same information-latency edge described in §5, on a different instrument. It isn't a separate, undocumented strategy; it's the same edge, manually extended to whatever market the same logic applied to on a given day.

On the remainder. The final +$5.83 (0.8%) is one additional settled trade, placed one day outside this paper's declared sample window (2026-06-15 → 2026-07-19) — real, realized P&L that landed in the account's total balance, but falling outside the date range every other table and breakdown in this paper is scoped to. It doesn't affect any strategy-level figure elsewhere in this paper.

12 Statistical Validation

12.1 Monte Carlo test: is Strategy I's gap mean-reversion real, or noise?

Null hypothesis (H_0): the `gap` signal (§4.1) follows a pure, driftless random walk.

Calibration. Per-step increments of `gap` were computed from 182,760 consecutive same-player/same-game observations (median step ≈66 seconds): mean ≈0 (no drift), standard deviation **$\sigma = 0.0227$** per step.

Simulation. For each of the 27 Strategy I trades (from the 56 clean pre-kickoff exits, §9) with a resolvable gap reading at both entry and exit, 5,000 random-walk paths were simulated per trade — starting at that trade's actual entry gap, taking its actual number of observed steps — and the fraction ending with `|gap|` smaller than the starting `|gap|` recorded.

Result:

	OBSERVED	SIMULATED NULL (RANDOM WALK)
Narrowing rate	**63.0%** (17/27)	**15.1%**

A one-sample z-test gives **$z = 6.94$, $p < 0.000001$** — not explainable by chance under a volatility-matched random walk, and the strongest single piece of evidence in this paper for genuine, tradeable mean reversion in Strategy I's core signal.

The obvious objection: the null is unbounded, the real signal is not. Contract prices live in [0,1], so `gap` is bounded too, while the simulated random walk is free to wander past those limits. A bounded process starting at a large |gap| must revert eventually simply because it cannot escape — which would inflate the observed-versus-simulated difference rather than reflect genuine mean reversion. This objection is real and worth stating plainly. Two things bound how much it can explain. First, the effect size is very large: 63.0% against 15.1% is not a margin a boundary artifact plausibly manufactures at these gap magnitudes, where the median |gap| at entry sits far from either price limit. Second, the null's dispersion over a typical holding period ($\sigma = 0.0227$ per step, median step ≈66 seconds) is what drives the low 15.1% figure — the simulated walk usually ends *further* from zero because it has room to drift, not because it is being pushed outward. A reflecting-barrier null would raise the simulated narrowing rate somewhat and shrink the gap; it would not close it. Re-running this test with explicit barriers at the price bounds is the correct next refinement, and is listed in §14 as an open item rather than claimed as done.

12.2 Calibration test: does Strategy II beat the market's own pricing?

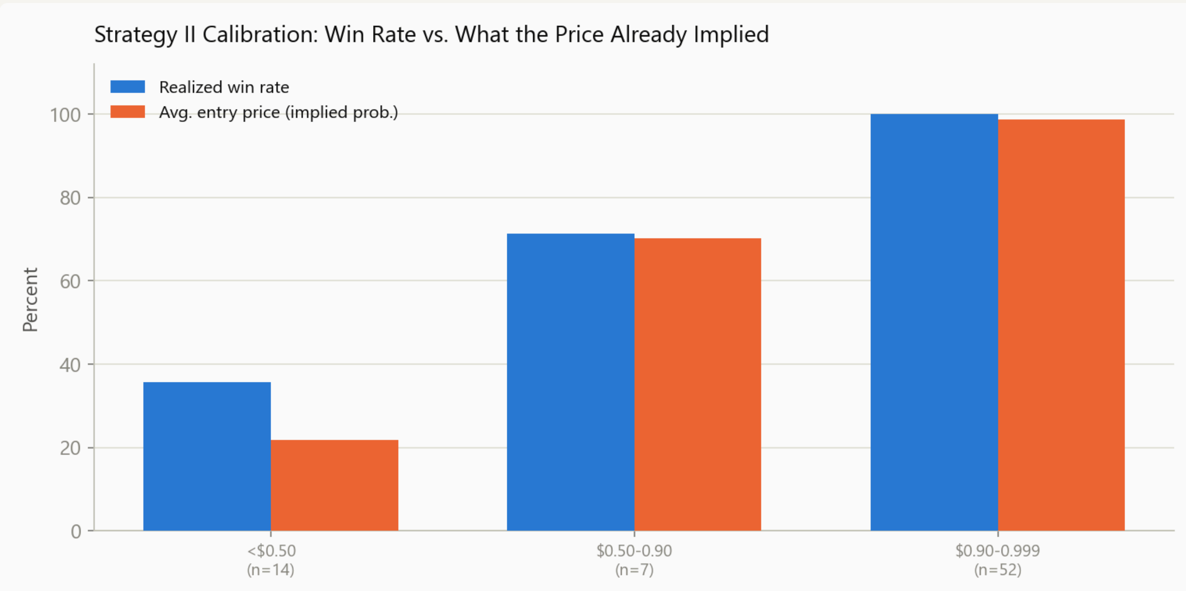
Positive side. Null hypothesis (H_0): the market was already efficiently pricing the contracts Strategy II's positive side bought.

Data. 73 positions — every `settlement_history` row (§9) with a matching average BUY-side entry price in `trade_history`, joined directly by `market_slug` rather than the JSON-parsed, tier-slug-regex matching this section originally used. That regex only recognized the goals/assists/goals+assists prop-tier slug pattern, so it silently excluded every resolved position outside it; joining on the real key instead pulls in the complete local population of clean, buy-and-hold-to-resolution positions across every market type the account traded — including the same speed-of-execution mechanism applied to non-prop markets (§11's futures, game-to-advance, and shot-prop bets), not just the 62 confirmed-goal entries (§5.2). Short-initiated positions are excluded automatically, not by hand: a `market_slug` with no BUY volume in `trade_history` (Strategy II's negative side, §5.3, which opens by selling) simply has no entry price to join against.

Result: realized win rate **84.9%** (62/73) against an average entry price of **81.2%** — $z = 0.81$, $p = 0.21$ — not statistically significant, and effectively unchanged from the narrower slice this replaces (the win rate is identical to the decimal; the entry price moved 80.4% → 81.2%), confirming the original conclusion wasn't an artifact of the population this test used to be restricted to.

By entry-price bucket (three wide groups so each has a meaningful sample size — the finer five-bucket split had two buckets with n=2 and n=4, too thin to interpret):

ENTRY PRICE BUCKET	N	WIN RATE	AVG. ENTRY PRICE
< $0.50	14	35.7%	$0.218
$0.50–$0.90	7	71.4%	$0.702
$0.90–$0.999	**52**	**100%**	**$0.987**



Realized win rate vs. average entry price, by bucket. Buckets track each other closely — the market was already priced fairly.

The dominant bucket by volume (52 of 73) resolved "Yes" 100% of the time against a 98.7% average entry price — exactly what a well-calibrated market predicts, not evidence of mispricing. **Strategy II's positive-side profit is real and mechanically reliable, but not evidence of beating the market's own pricing** — the edge is being fast enough to buy before the book reprices, not being smarter about the true probability once bought.

On the `<$0.50` bucket. It grew from n=13 to n=14 — one more real position (a $0.29 `game_to_advance` bet that lost), found only because the join no longer requires a goal-tier prop slug to match. That's the honest ceiling: this is now the complete local population of buy-and-hold positions priced under $0.50 across the account's full 35-day history, not a partial slice that might be hiding more. A larger sample here requires more trading history, not better mining of what's already collected. At 35.7% realized (5/14) against a 21.8% implied probability, an exact binomial test (appropriate at this sample size, unlike the normal approximation used above) against the null of efficient pricing gives **$P(\geq 5 \text{ wins of } 14 \mid p=21.8\%) = 0.173$** — still not significant at any conventional threshold, but now a precise number against a confirmed-complete population rather than a hand-wave about a small n. The direction is consistent with genuine edge; the evidence still isn't strong enough to claim it.

Negative side. Null hypothesis (H_0): the market was already efficiently pricing the contracts Strategy II's negative side sold.

Data. 102 positions — every `settlement_history` row tagged `strategy_ii_negative` with `market_type` in {goals, assists, goals+assists} (§5.3's exact population), joined to `trade_history`'s average SELL-side price by `market_slug`. The analog of "entry price" here is the implied probability of the account's actual bet — a **"No" price** of `1 - average Yes-sell price`, since selling Yes at a low price is how "buying No" executes on this exchange (§5.3, §15). Every position in this population has real SELL volume by construction (that's what makes it the negative side); the win/loss/flat split uses the same ±$0.005 threshold as every other closed-trade classification in this paper (§5.2, §5.3), not a new convention invented for this test.

Result: realized win rate **89.2%** (91/102, matching §5.3's published split exactly — 91 wins, 5 losses, 6 flat) against an average implied "No" price of **95.6%** — $z = -3.15$, $p = 0.0016$ (two-sided) — statistically significant, and in the opposite direction from the positive side: the negative side's realized results are **worse** than what its own entry prices implied, not indistinguishable from them.

By entry-price bucket:

"NO" PRICE BUCKET	N	WIN RATE	AVG. IMPLIED "NO" PRICE
< $0.50	0	—	—
$0.50–$0.90	13	30.8%	$0.764
$0.90–$0.999	**89**	**97.8%**	**$0.984**

No position was ever entered below 50% implied confidence — unlike the positive side, this side's trigger (a confirmed substitution or full-time whistle, §5.3) doesn't fire until the account is already fairly sure. The dominant bucket (89 of 102, 87% of the population) is calibrated almost exactly as well as the positive side's dominant bucket: 97.8% realized against a 98.4% implied price. **The entire shortfall lives in the minority $0.50–$0.90 bucket** (n=13): a 30.8% realized win rate against a 76.4% implied one. Two positions drive it — Charles De Ketelaere (−$73.47) and Jovo Lukić (−$46.38), §5.3 — genuine signal misses (the confirmed non-event didn't hold) rather than execution artifacts, together with three smaller losses and several flat/voided outcomes that also cluster in this same lower-confidence tier.

Strategy II's negative side is not calibrated as cleanly as its positive side. The dominant, highest-confidence majority of both sides prices fairly. But where the positive side's `<$0.50` bucket only *hinted* at something (n=14, not significant, §12.2 above), the negative side's $0.50–$0.90 bucket is both large enough and skewed enough to be a real, Bonferroni-surviving finding (§12.3): when this account's "confirmed non-event" signal wasn't yet fully certain, it underperformed its own pricing by a wide and statistically real margin, not just a fluky one.

12.3 Multiple comparisons and in-sample validity

Multiple comparisons. This paper runs three primary formal hypothesis tests (§12.1's gap-narrowing test, and §12.2's positive- and negative-side calibration tests) plus several descriptive bucket breakdowns (§8, §12.2). Bonferroni-correcting all three primary tests (threshold 0.05/3 = 0.0167) changes no conclusion: §12.1 (p<0.000001) and §12.2's negative-side result (p=0.0016) both clear the corrected threshold comfortably; §12.2's positive-side result (p=0.21) remains non-significant either way. The descriptive bucket breakdowns are explicitly *not* treated as independent hypothesis tests anywhere in this paper.

In-sample validity. Strategy I's `formula` / `gap` signal was both *designed* and *evaluated* against the same 35-day dataset — there's no held-out, out-of-sample period tested against. This is a real limitation: a signal discovered by looking at data and then "confirmed" by looking at the same data again is the classic setup for overstating a result (data-snooping bias). Partially mitigating it: the signal's mathematical form (§4.1's inclusion–exclusion identity) was derived from first principles before this dataset existed, not fitted to it — but the decision to trade on it, and the entry/exit behavior tested in §12.1, were both shaped by watching this same tournament unfold.

A genuine out-of-sample test needs a second tournament with the same market structure, and a complete, live census of every soccer competition Polymarket US lists finds none currently has one. Strategy I's signal depends on three parallel per-player markets — goals, assists, and goals+assists on the same game — the exact structure that makes the `formula` / `gap` identity computable at all. `GET /v2/sports` returns Polymarket US's own authoritative list of every league it operates, per sport — not a guessed handful of slugs: soccer alone spans **70 leagues** besides the World Cup, from the EPL, UCL, and MLS down to regional second divisions and continental cups. Discovering every confirmed game across **all 70** and checking each of their **141 currently-upcoming games** directly against the live API shows **zero** player goal/assist prop markets anywhere outside the World Cup — every one of the 141 offers only team-level markets (winner, spread, total, exact score, both-teams-to-score). This isn't a sampled check; it's a complete census of every upcoming soccer game on the platform as of 2026-08-08.

That sharpens the follow-up rather than closing it off. The collector (§3, §9) now discovers events across the World Cup plus all 70 other soccer leagues, with the league list itself pulled live from `GET /v2/sports` (`find_soccer_league_slugs`) rather than hand-maintained — a league Polymarket adds or removes is picked up automatically, with no code change. This full scan runs once daily rather than on every poll (a ~70-league sweep and a 60-second polling cadence don't mix), and already found real, immediate value beyond the props question: **1,337 confirmed games newly logged to `discovered_games` in a single run**, growing the local dataset from 601 to 1,938 games — a real, substantially larger inventory of tracked soccer fixtures than existed before this check, even though none of the new ones carry player props yet. Data collection for a real out-of-sample replication starts automatically the moment any league lists a player goal/assist market again, with no further setup required.

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13 Capital Structure, Turnover, and Risk Metrics

13.1 Turnover

The account's $1,200 in net contributed capital generated **$27,587 in gross buy volume and $33,736 in gross sell proceeds** across 879 logged trade fills — roughly **23x the contributed capital** in cumulative buy volume alone. Top trading days logged 40–71 fills in a single day (Appendix B) — a significant part of how $1,200 became $691.58 in realized P&L (57.6% return, all-in) inside 35 days.

13.2 Equity curve and maximum drawdown

A cumulative equity curve from all 356 P&L-realizing events (192 Strategy I sells plus 164 Strategy II settlements, ordered by timestamp) gives a real, computed maximum drawdown. Daily closing values (cumulative realized P&L on the closed-trade-and-resolution series, excluding the ~$63 non-prop and $5.83 remainder components of §11, so it understates the account's true final balance slightly):

Cumulative realized P&L, daily close, across both strategies.

METRIC	VALUE
Peak cumulative realized P&L reached	$623.03
Maximum drawdown	**$73.94 (11.9% of peak)**
Drawdown episode	$386.51 → $312.57, around 2026-07-10
Recovery	Full — the equity curve ends **at** its all-time peak ($623.03)

The drawdown lands on 2026-07-10 — the same date as the Spain–Belgium game containing the largest overfill-related losses (§5.3) — and fully recovers afterward; the series ends at its own historical maximum. The early dip on 06-24 (cumulative −$57.21) is the account's worst relative point in the entire sample — every figure in this paper from 06-26 onward is a new all-time high, not a single lucky run followed by stagnation.

Decomposed into day-over-day changes rather than shown cumulatively, the same series is **19 net-positive days against 6 net-negative days** (3 days had fills but nothing closed or settled):

The single worst day, 06-24 (−$48.09), is more than offset within 48 hours by 06-25 and 06-26 combined (+$88.28); the 07-10 drawdown day (−$38.92) is similarly a single red bar inside an otherwise green second half of the sample, not the start of a losing stretch.

13.3 A time-weighted daily NAV, and the Sharpe and Calmar ratios it supports

A Sharpe ratio requires a consistent per-period return series with a well-defined volatility to normalize against. This account doesn't have one by default: position sizes were highly irregular (§10's $17.98 average vs. $214.45 maximum), capital was recycled multiple times within a single day, and returns are event-driven (clustered around match days, §13.1) rather than time-driven. The fix is a real, mark-to-market NAV series rather than the realized-only equity curve in §13.2:

> **NAV(t) = net deposits so far + cumulative realized P&L (same series as §13.2) + unrealized P&L on currently-open positions.**

Net deposits come from the local `cash_activity` table (§6) — four completed deposits and one withdrawal, each timestamped to when the funds became usable rather than when ACH settlement later confirmed them (the account's first trade fires 27 minutes after the first deposit request, two days before that deposit's official completion timestamp — funds are usable immediately). Unrealized P&L marks every currently-open position against `position_price_history` at each day's close, with positions tracked per-contract (buy/sell **and** short-sell/cover, since Strategy II's negative side, §5.3, opens positions by selling before ever buying) using `price` and `qty` only — average-cost accounting, same convention as everywhere else in this paper (§7).

Because capital arrived in four installments rather than as a single day-one lump sum, daily returns are computed with the **Modified Dietz** method, which excludes each day's deposit or withdrawal from that day's return so a cash *contribution* is never counted as a trading *gain*. The result is a genuine time-weighted return

(TWR) series, distinct from the §13.2 curve in an important way: §13.2 sums dollar P&L over a fixed $1,200 base, while the TWR series compounds each period's return against whatever capital was actually deployed *at that time* — capital that was $100–$300 for the first ten days, not $1,200. The same early dollar gains therefore carry more weight in the TWR series, which is why it finishes higher:

METRIC	VALUE
TWR total return (35 days, compounded, cash-flow-neutral)	**+98.3%**
Daily returns (n)	31
Mean daily return / stdev	+2.37% / 5.64%
Sharpe ratio, daily	**0.42**
Sharpe ratio, annualized	**not reported — see below**
Max NAV drawdown	**6.6%**, 07-02 → 07-03
Calmar-style ratio (raw 35-day return ÷ max NAV drawdown)	**14.9**

Why the annualized Sharpe is deliberately not reported. Scaling this daily figure by √365 yields roughly 8.0, and an earlier draft of this paper printed that number. It should not have. With **31 daily observations** drawn from a single bounded event, the sampling error on a Sharpe estimate is severe — the approximate standard error of a Sharpe estimate is $\sqrt{((1+S^2/2)/n)}$, which here puts a 95% confidence interval on the *daily* figure of roughly −0.06 to +0.90. Annualizing multiplies that entire interval by 19.1, producing a range wide enough to be uninformative while presenting as precision. The daily value is the honest one; the annualized value is arithmetic performed on too little data. The same reasoning already governs the Calmar figure below, which is reported for the raw 35-day period and deliberately not annualized: compounding a 35-day return out to a full year (`(1+r)^(365/35)`) explodes past 100,000%. Readers who want the annualized number can compute it; this paper declines to headline it.

Robustness check. Restricting the same calculation to 2026-06-26 onward — after all $1,200 was deposited, removing the low-capital ramp-up entirely — gives

Sharpe 0.57 daily on n=23 returns and a Calmar-style ratio of 7.3. The Sharpe estimate is *higher*, not lower, once the ramp-up is excluded — it isn't propping up the headline figure — while Calmar roughly halves, since ten days of uninterrupted early gains leave the numerator with less room to compound. Both estimates rest on a small sample (23–31 daily observations) and should be read as short-sample estimates, not a claim about this strategy's long-run risk-adjusted return.

The NAV drawdown lands on a different date than the realized-only drawdown. §13.2's $73.94 drawdown centers on 2026-07-10; the NAV series' worst point is 07-02 → 07-03, and 07-10 barely registers on it. Some of what became a realized loss on 07-10 was already priced into open positions' mark-to-market value in the days before it settled — a real difference between "loss when it closed" and "loss when the market started pricing it in," visible only once unrealized P&L is tracked at all.

13.4 Capacity

A strategy's *capacity* — how much capital it can absorb before its own activity erodes its edge — is capacity-constrained here by construction. §8 measured top-of-book depth on the core goals+assists instrument at 1,406–9,762 shares (roughly $700–$5,000 notional); §10's average position size was $17.98. Deploying meaningfully more capital would mean either walking through multiple price levels on entry (eroding the mispricing being captured) or spreading across more players/games than one person can manually monitor within the latency window Strategy II depends on (§5). The strategy is real and repeatable at this scale; nothing here should be read as evidence it scales to institutional size without redesigning the execution layer (§16).

A third capacity constraint, and the clearest one in the data: capital grew exactly as opportunity shrank. Tournament game count by week (`discovered_games`) tapers sharply as the format progresses — 11, 28, 34, 18, 9, 4 games/week — while the account's cumulative realized P&L, and so its available bankroll, grew monotonically the entire time: roughly $1,190 → $1,275 → $1,514 → $1,651 → $1,823 across those same six weeks. The account had the *least* capital when opportunity was highest, and the *most* when opportunity was lowest.

This is the concrete reason this strategy was run against a single, bounded event rather than continuously: it was sized around a hypothesis specific to the World Cup's retail-attention surge (§1), not tested as an always-on strategy. Staggering exposure within a single day — redeploying capital freed by one game's resolution into a later same-day kickoff — partially mitigates this, but not across the tournament-wide capital/opportunity mismatch above.

A second, independent capacity constraint sits on the settlement side. Comparing each prop resolution's confirmation timestamp against the last trade on that market shows a **median 38-minute, mean 44-minute gap** between last trading activity and settlement confirmation (n=152 matched resolutions, 10th/90th percentile 3.4–102.1 minutes) — capital tied up for a real window after a game-ending event before it's freed back up as spendable balance. This matters because tournament scheduling clusters games: **25 of 35 trading days had two or more games**, several with near-simultaneous kickoffs (Appendix B), meaning capital tied up in one game's pending settlement could be unavailable for a second game's full-time event around the same time — a real, capital-side capacity ceiling this account hit in practice (§13.5).

13.5 Position sizing

Strategy I used a defined rule: fixed-fractional, with a hard risk cap enforced at kickoff.

PARAMETER	RULE
Base sizing	Fixed fraction of available capital per position
Entry mechanism	Built incrementally across multiple fills at improving prices (§4.3) — actual size could grow past the intended fraction before the entry finished
Hard cap	**5% of capital**, checked at kickoff
Enforcement	Active, below-market “panic sell” if the cap would otherwise be breached

Because a position was built incrementally across multiple fills (§4.3), its actual size could grow past the intended fraction before the entry finished; if a position's size at kickoff would push total portfolio risk above 5% of capital, it was cut down via the panic sell above — a second, deliberate reason (beyond §4.4's fill risk) some of the 136 delayed exits underperformed the clean 56.

Strategy II used a different rule, on both sides: breadth over size, not a probability-weighted stake.

PARAMETER	RULE
Base sizing	Capital spread across all still-unresolved candidate positions at the moment of the trigger (§5.3), not concentrated on one
Sizing basis	Liquidity allocation — how much capital can be deployed before running out (§13.4), not a probability-edge calculation
Kelly criterion applicable?	**No** — §12.2 finds realized win rate statistically indistinguishable from entry price (z=0.81, p=0.21), so there is no demonstrated $p > P$ edge to size against

Why Kelly criterion doesn't apply here. Kelly-style sizing, using the entry price as the probability estimate, is a natural-looking candidate for Strategy II, but it doesn't survive scrutiny. Kelly only justifies sizing up when true probability p exceeds the market's implied probability P (the entry price) — at $p = P$ exactly, the optimal stake is zero. §12.2 found no demonstrated $p > P$ edge, and using the entry price as the probability estimate is circular by construction. More fundamentally, once a goal is confirmed or a player can no longer contribute, the *true* probability isn't genuinely "98%" — it's much closer to 100%, with price merely lagging a fact that's already true. The edge is a **timing** edge (buying before the book reflects known reality), not a probability-estimation edge in the sense Kelly is built for. The right sizing question isn't "how much edge do I have over the market's probability estimate" — it's "how much capital can I deploy before I run out, given execution and settlement constraints" (§13.4), a liquidity problem. The capital-spread-across-many-positions heuristic actually used (§5.3) is a reasonable practical answer to that liquidity question; Kelly criterion is not the right tool here.

Formalizing the liquidity-allocation rule. "Spread across all still-unresolved positions" is a real heuristic, but not a formal one — it says *how many* positions to spread across, not *how much* to put in each. The natural formalization, given the settlement-lag capacity constraint already established (§13.4): size each simultaneously-triggered candidate **inversely to its expected time until capital is freed**,

```
w_i = (1 / T_i) / Σₒ (1 / T_j)
```

for the set of candidates j competing for capital at the same moment, where T_i is the expected minutes from entry to settlement — a position expected to free its capital quickly earns a larger allocation than one expected to sit tied up for hours, for the same reason a market maker sizes down its slowest-turning inventory: capital that comes back sooner can be redeployed into the *next* opportunity sooner, which is the actual scarce resource here (§13.4), not probability edge.

T_i has a real, measurable empirical distribution: matching each of the 164 confirmed-event positions' entry fill to its settlement timestamp, excluding 6 positions where the gap exceeds 24 hours (inconsistent with a mechanism that operates within a single, roughly two-hour match, §5.1 — these read as older speculative entries later swept into this side by the position-sign-based strategy tag, §9, not genuine confirmed-event triggers) gives **n=158, median 43.6 minutes (IQR 17.6–81.2 min)** — closely matching §13.4's independently-derived 38–44 minute settlement lag, a useful cross-check that this T_i measure is capturing the same real constraint from a different angle.

Was the actual heuristic already doing this, even informally? No. Correlating each position's actual dollar allocation against its $1/T_i$ weight, within each day's set of simultaneously-triggered candidates (n=157 position-day pairs across 23 multi-position days), gives a Spearman **ϱ=0.078** — indistinguishable from zero. Raw position cost against raw T_i, ignoring day-grouping entirely, gives ϱ=−0.032. “Spread across all still-unresolved positions” was, in practice, close to indifferent to how quickly each one would free its capital — closer to an equal-weight split than a velocity-weighted one. Formalizing the rule this way would be a real behavioral change, not a restatement of what already happened.

A naive backtest of this rule is not credible, and it's worth showing why. Reallocating each day's actual capital by the $1/T\_i$ weights above, then applying each position's own realized return-per-dollar to its new (larger or smaller) allocation, implies an aggregate return of **+34.5%** against the account's realized **+1.8%** on this population — an implausible 20x improvement. The mechanism is a real backtesting trap, not a real edge: a handful of dust-sized, fast-resolving positions posted large *percentage* returns on trivial dollar amounts (§10's fee-residual pattern, §11), and naively scaling those percentages up to realistic dollar sizes ignores the exact capacity constraint this paper already established (§8, §13.4) — the liquidity to fill a $0.01 position at 100x its actual size simply wasn't there. This number is reported to show it was checked and rejected, not as a finding.

A capacity-aware benefit is directly quantifiable, though, and appears in §13.6's exit-rule backtest below: the same T_i data underlies both questions, since exiting a position earlier and sizing it by how fast it frees capital are two sides of the same liquidity-management problem. The correlation result above is the honest empirical takeaway for sizing specifically: the formal rule is well-posed and grounded in real, cross-validated data, but this account's actual behavior didn't already implement it, even approximately.

13.6 Hold-to-resolution vs. early exit: a real, unexploited optimization

Given a confirmed-winning Strategy II position, does it make sense to hold to full $1.00 settlement, or exit early near $0.99 to recycle capital faster given §13.4's settlement lag? The stated reason for "always hold" — that the $0.98–$0.99 margin is too thin to bother capturing (§5.1) — is real and confirmed against the trade record, but only half the analysis.

In practice, this account never sold a near-certain position early — zero of 192 Strategy I closed trades (§7) were sold at a price ≥ $0.90, and Strategy II never sells on either side by design. There's no historical "sold early" comparison group, because the early-exit option was simply never used.

What the data does show: of the 164 Strategy II held-to-resolution positions, **65.2% (107 of 164) resolved on a day with 2 or more games running** (average 3.07 games that same day) — the exact condition §13.4 identifies as capital-constrained. The “always hold” behavior was individually rational (the per-position margin really is thin), but it was disproportionately applied on precisely the days when tying up capital for that thin marginal gain had the highest *aggregate* opportunity cost. This doesn't prove a specific dollar amount was left on the table, but it's a real pattern: **the per-trade logic (“this cent isn't worth selling for”) and the portfolio-level logic (“freeing this capital sooner might matter a lot on a busy day”) can both be correct and still point to different actions.** A simple rule conditioned on day-level busyness — sell shortly after the trigger on multi-game days, hold to full settlement on quiet ones — is the natural reconciliation, and can now be backtested like-for-like against the actual hold-everywhere policy using the same *T_i* (entry-to-settlement) data built for §13.5's sizing rule.

The backtest. Of the 158 confirmed-event positions with a non-outlier *T_i* (§13.5), 101 resolved on a multi-game day — this rule's target population — tying up **491.4 capital-dollar-days** to realize **$175.24** in margin. Exiting each of these near its own entry price roughly 3 minutes after the trigger, instead of holding for the full settlement wait, would forfeit essentially all of that $175.24 — the whole point of “always hold” is that the margin *is* the reward for waiting, so an immediate exit captures none of it — while freeing **465.7 of those capital-dollar-days** for redeployment (a dollar-weighted average wait of 54.4 minutes per position, cut to ~3).

Whether that trade is worth making depends on what the freed capital could have earned redeployed elsewhere, which this dataset can't observe directly. Using the account's own realized rate across the same non-outlier population as an empirical benchmark — **$331.25 realized on 716.1 capital-dollar-days, or 46.3% per capital-dollar-day** (a raw ratio over 158 short-lived positions, not an annualizable rate — the same compounding-explosion caution as §13.3's Calmar figure applies if this number is extrapolated) — the 465.7 freed capital-dollar-days would need to earn just **37.6% of that same per-capital-dollar-day rate** to break even against the

$175.24 forfeited, and **81.3% of it** to match the account's own typical rate exactly. At the account's own realized rate, the freed capital implies **$215.43** in redeployed P&L against $175.24 given up — **a net +$40.19**, a modest, real, and directionally consistent result rather than a large or definitive one. The result depends on an assumption (freed capital finds another opportunity at the account's own typical rate) this dataset can't independently verify, but the breakeven bar is low enough — recovering just over a third of the account's own typical rate — that the qualitative case from §13.4 (idle capital on the busiest days had the highest opportunity cost) survives a real, capacity-aware quantitative check, not just a plausibility argument.

3 additional supporting charts for this section are in Appendix E.

14 Risk Analysis and Limitations

- **Sample size.** 192 Strategy I trades and 164 Strategy II settlements support the §12 tests, but the `<$0.50` calibration bucket — now n=14, the complete local population rather than a partial slice (§12.2) — is still explicitly underpowered.
- **In-sample validity.** Strategy I's signal was designed and tested on the same dataset, with no out-of-sample holdout yet — the single biggest open methodological question in this paper. §12.3's complete census of all 70 other soccer leagues Polymarket US lists confirms none currently has the market structure this would require, and the collector now discovers events across all of them, live rather than hand-listed, so real out-of-sample data collection starts the moment one does.
- **Trade clustering / effective sample size.** The 192 Strategy I trades span only 28 distinct games, and the top 5 alone account for 51% of all trades. Treating n=192 as 192 independent observations (as the descriptive win-rate statistics in §10 implicitly do) overstates precision relative to the true number of independent *events*. §12's formal tests sidestep this by testing the signal's behavior rather than each trade independently, but it's a real limitation of the simpler win-rate figures elsewhere.
- **Model risk on the core signal** (§4.2).

- **Adverse selection risk cuts both ways** (§4.2).
- **Execution provenance.** Strategy I's live order-placement logic is a reconstruction, not sourced code (§4.3).
- **Fill risk is real and quantified, not hypothetical.** 136 of 192 Strategy I positions (71%) didn't exit before kickoff as designed, and that population's win rate (78.7%) and average return were measurably worse than the 56 that did (94.6%) (§4.4, §9).
- **Manual-execution latency in Strategy II** (§5.1).
- **The reaction-time distribution rests on 7 independently-verified goals so far** (§5.1, mean 91.0s, median 96.8s) — real and not circular, but still a small slice of all 164 confirmed-event positions, and limited to first-half goals until second-half timing has a real independent source for the actual halftime-break length.
- **Strategy II's negative side, now covered by an extended §12.2 calibration test, shows a real, statistically significant shortfall against its own pricing** ($z=-3.15$, $p=0.0016$, survives Bonferroni correction across all three of this paper's formal tests, §12.3) — concentrated entirely in the minority of positions entered below 90% implied confidence ($n=13$ of 102), where two genuine signal misses (Charles De Ketelaere −$73.47, Jovo Lukić −$46.38, §5.3) account for most of it. The dominant, high-confidence majority (89 of 102 positions) is calibrated about as well as the positive side.
- **A $5.83 (0.8%) remainder in the overall P&L reconciliation** (§11) remains unattributed to a specific component, though its size and sign are consistent with real trading fees rather than any missing activity.
- **A ~$1 (0.4%) tie-break sensitivity** among a handful of literally simultaneous-timestamp fills at different prices, where the true execution order can't be recovered from the data (§7).

What Would Falsify This

Limitations describe what is unknown; this section states what evidence would make the paper's claims wrong. Each is a pre-commitment, written before the data that could settle it exists.

1. **The mean-reversion result (§12.1) would be falsified** if the same Monte Carlo test, applied to a second tournament's clean pre-kickoff trades, returned a narrowing rate near the null's 15.1% rather than near 63.0%. That outcome would mean the signal was tournament-specific noise fitted in-sample, exactly as §12.3 warns is possible. A rate materially between the two would indicate a real but weaker effect than reported here.
2. **It would also be substantially weakened** if re-running the null with reflecting barriers at the price bounds (§12.1) moved the simulated narrowing rate close to the observed 63.0%. That would show the result was a boundary artifact rather than genuine reversion.
3. **The structural finding (§8) would be falsified** if the 28.81% logical-floor violation rate collapsed toward zero once other venues, other sports, or a later period were measured — indicating a transient artifact of one product during one event rather than a property of thinly-quoted compound contracts generally.
4. **The capacity claim (§13.4) would be falsified** in the favorable direction if deploying materially more capital left realized spread capture unchanged. This paper predicts it would not: at measured top-of-book depth of 1,406–9,762 shares against an average $17.98 position, size should erode the very mispricing being captured.
5. **Strategy II's positive-side conclusion would be overturned** if a larger sample of sub-$0.50 entries (currently n=14, the complete local population) showed a win rate durably above its implied probability. The direction is currently consistent with edge (35.7% realized vs. 21.8% implied, exact binomial p=0.173); the sample cannot support the claim.

None of these can be resolved by further analysis of the existing dataset. Each requires new data, and §12.3 documents the collection infrastructure already in place to capture it automatically.

15 System Notes

- **Rate-limited API handling**, with a documented backoff contract for both the public gateway and the authenticated portfolio API.
- **Explicit read/write boundary.** `TradingClient` is GET-only by construction — `get_balances` (§6) extends that same read-only client rather than introducing a new trust boundary.
- **Self-generating SQL views**, re-executed against live base tables on every read, so signal state (`live_formula_gap`) is always current without a separate materialization step.
- **Confirmed directly against the live API**: Polymarket US runs one order book per question — "Yes" and "No" are two sides of the same instrument (`marketSides` in `get_market_by_slug`), not independently tradeable tokens. This settles what "buying No" mechanically is (a `SELL` of "Yes," §5.3) and rules out a classic complementary-token arbitrage strategy for this product (§16).

16 Conclusion and Future Work

Two forces were monetized against the same underlying inefficiency — a retail-heavy, thinly-quoted niche derivatives market during a high-attention sporting event: patience, relative-value mean reversion between correlated instruments, statistically validated; and speed, latency on public information events, traded on both sides of a single confirmed-event mechanism. Strategy II's positive side is real but mechanical — a calibration test finds no evidence it beats the market's own pricing once bought, the honest result for a strategy about being fast rather than smarter. Its negative side is the inverse honest result: a calibration test finds it real but worse than its own pricing implied, entirely within the minority of trades entered before full certainty — a genuine, statistically significant finding, just not a favorable one. Strategy I's edge, by contrast, is this paper's one favorable statistical claim: its core signal narrows toward fair value far more often than chance would predict ($p<0.000001$). The account converted $1,200 in contributed capital into $691.58 in fully realized trading P&L (57.6% all-in) over 35 days, fully reconciled against the account's live transaction history.

The scaling constraint is structural, not incidental. The clearest single fact in this dataset about whether the approach generalizes is that capital grew exactly as opportunity shrank: tournament game count fell 11 → 28 → 34 → 18 → 9 → 4 per week as the bracket narrowed, while the bankroll rose monotonically from roughly $1,190 to $1,823 across those same weeks (§13.4). The account held the least capital when opportunity was greatest and the most when it was smallest. Combined with measured top-of-book depth of 1,406–9,762 shares against an average position of $17.98, and a median 38-minute lag between a game-ending event and settlement freeing capital, this strategy is capacity-bound by construction. Nothing here should be read as evidence it survives institutional size without a redesigned execution layer — and the honest version of that claim is that the redesign, not the signal, is the hard part.

Concrete next step

1. **Run the out-of-sample Monte Carlo replication** (§12.1's test, applied to a second tournament) the moment one becomes available. This is no longer a design question — §12.3 confirms the collector already discovers events across the World Cup and all 70 other soccer leagues Polymarket US lists, and starts collecting automatically as soon as any of them lists a player goal/assist market again — but it does need real time to pass and a real market to appear, neither of which a single analysis session can manufacture.

> *(A complementary YES/NO-token arbitrage strategy — buying both sides when they sum below $1.00 and redeeming for a guaranteed $1.00, a pattern seen in comparable open-source Polymarket trading systems — was considered and ruled out for this product:* `get_market_by_slug` *confirms Polymarket US runs one order book per question, with "Yes" and "No" as two sides of the same instrument rather than independently tradeable tokens, so the classic complementary-arb structure doesn't apply here.)*

Appendix A — Glossary

CLOB

Central limit order book.

BBO

Best bid/offer.

Spread

Best offer minus best bid.

Depth

Size resting at a given price level.

Passive / aggressive fill

Passive: your resting order got hit. Aggressive: you crossed the spread.

Fill risk

The chance a resting limit order does not execute within the intended window, forcing a position to be carried longer than planned. Quantified for Strategy I in §4.4, §9.

Settlement / redemption

Terminal resolution to $1.00/$0.00 — distinct from an active SELL.

Calibration

A price is calibrated if contracts priced at p resolve “Yes” ≈p of the time.

Inclusion–exclusion

`P(A∪B) = P(A) + P(B) − P(A∩B)`, used here under an independence approximation.

Mean reversion

Tendency of a price/signal to move back toward a reference level. Tested formally in §12.1. The “patience” mechanism (§1).

Latency arbitrage

Trading on a lag between an information event and the market's repricing. The “speed” mechanism (§1).

Average-cost (pooled) accounting

Single running `(qty, cost)` per position; every sell prices at the current pooled average, reducing both proportionally. Order-independent by construction (§7).

Monte Carlo null test

Simulating many paths under a “no real effect” model, calibrated from real data, to see whether an observed result is distinguishable from chance.

z-test (one-sample proportion)

Tests whether an observed proportion differs from a hypothesized one by more than sampling noise predicts.

Endgame arbitrage

A named strategy in comparable prediction-market trading systems: buying whichever side of a market has become near-certain shortly before resolution and holding to settlement. Covers this paper's Strategy II in full — both the positive (“Yes”-side, §5.2) and negative (“No”-side, §5.3) applications.

NAV (net asset value)

Cash plus the current mark-to-market value of open positions — the account's true value at a point in time, vs. a realized-only P&L curve, which ignores positions still open. Built in §13.3.

Time-weighted return (TWR)

A return series compounded period-over-period, isolating trading performance from the size and timing of deposits/withdrawals. Contrast with a dollar return on a fixed capital base (§13.2's equity curve).

Modified Dietz method

A standard way to compute a period's TWR when cash flows land mid-period: the flow is excluded from that period's numerator so a deposit is never counted as a trading gain. Used in §13.3.

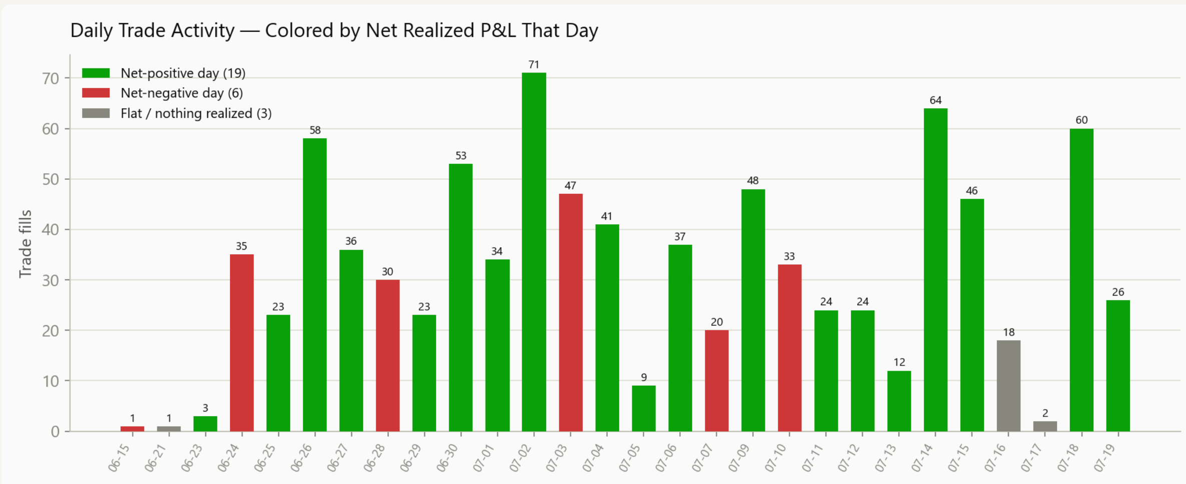
Sharpe ratio

Mean return over its own standard deviation, most often annualized by a scale factor (√252 for equity markets' trading days; √365 is used in §13.3, since this account trades every calendar day).

Calmar ratio

Return divided by max drawdown — a return-per-unit-of-pain measure. Reported for the raw 35-day period in §13.3, not annualized (annualizing a 35-day return to a full year is a large extrapolation).

Appendix B — Daily Trade Activity



Fills per trading day, colored by whether that day's realized P&L was net-positive, net-negative, or flat.

DATE	FILLS	DATE	FILLS	DATE	FILLS
06-15	1	06-29	23	07-11	24
06-21	1	06-30	53	07-12	24
06-23	3	07-01	34	07-13	12
06-24	35	07-02	71	07-14	64
06-25	23	07-03	47	07-15	46
06-26	58	07-04	41	07-16	18
06-27	36	07-05	9	07-17	2
06-28	30	07-06	37	07-18	60
		07-07	20	07-19	26
		07-09	48		
		07-10	33		

Appendix C — Related Systems Reviewed for This Paper

Reviewed while researching whether other strategies could be added to this project:

- `warproxxx/poly-maker` — a two-sided Polymarket market-making bot. Fair value computed as a depth-weighted microprice off the live book, nudged by an EWMA of signed trade flow; inventory skew shifts quotes asymmetrically as position size grows; five operational regimes (QUIET/TRENDING/EVENT/REDUCE_ONLY/HALTED) govern risk posture. Cited in §8 as a more rigorous, formalized version of the spread/depth dynamics this paper measures empirically but doesn't yet trade directly.
- `casatrick/polymarket-arbitrage-bot-python` — five strategies: intra-market arbitrage (YES+NO < $1.00), combinatorial arbitrage (multi-outcome markets), cross-platform arbitrage (vs. spot crypto prices), **endgame arbitrage** (the named equivalent of this paper's Strategy II, both sides, §5), and technical-indicator momentum/mean-reversion (z-score/RSI-based — a less principled analogue of this paper's Strategy I, which uses a probability-model signal rather than a purely technical one).
- `octavi42/prediction-market-maker` — inventory-skew case study (Paradigm Prediction Market Challenge, 2nd place); illustrates why skew-free market-making is fragile.

Appendix D — Academic References

Cited for terminology and concepts this paper uses but did not originate:

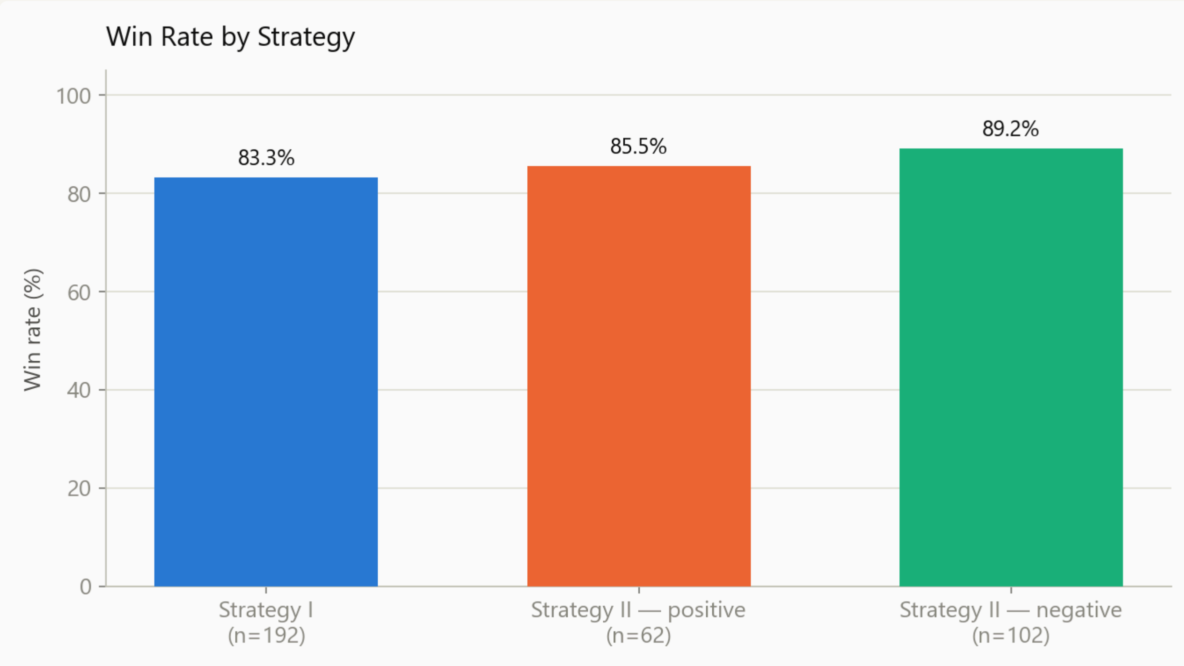
- Glosten, L. R., & Milgrom, P. R. (1985). *Bid, Ask and Transaction Prices in a Specialist Market with Heterogeneously Informed Traders.* Journal of Financial Economics, 14(1), 71–100. — Origin of the result that bid-ask spreads compensate market makers for adverse selection against informed counterparties; cited in §4.2.

- Kyle, A. S. (1985). *Continuous Auctions and Insider Trading.* Econometrica, 53(6), 1315–1335. — Origin of "Kyle's lambda," price impact per unit of order flow as a measure of informed-trading intensity; cited alongside Glosten-Milgrom in §4.2.
- Artzner, P., Delbaen, F., Eber, J.-M., & Heath, D. (1999). *Coherent Measures of Risk.* Mathematical Finance, 9(3), 203–228. — Standard reference for the properties a risk measure should satisfy; background for §13.3's Sharpe and Calmar ratios and the drawdown measures throughout this paper.
- Demsetz, H. (1968). *The Cost of Transacting.* Quarterly Journal of Economics, 82(1), 33–53. — Origin of the "immediacy" explanation for bid-ask spreads: sellers rationally accept a worse price in exchange for transacting now rather than waiting; cited in §5.5 as the complementary reason counterparties transact with this account's Strategy II positions.

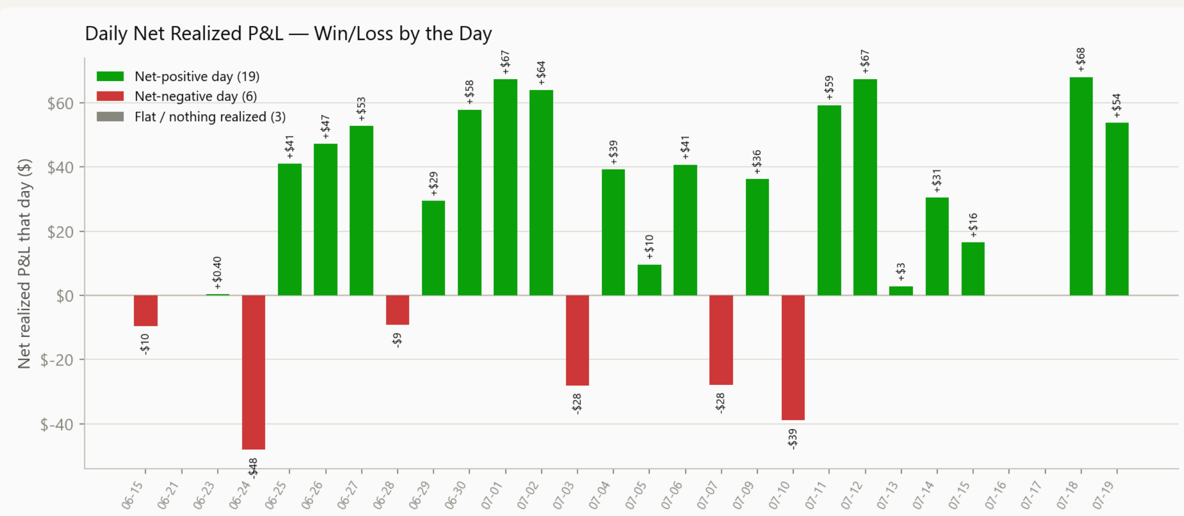
This paper doesn't claim methodological novelty relative to this literature — the contribution here is empirical (a real, audited trading record) rather than theoretical, and these references let a technical reader locate the underlying theory this paper's informal language (§2, §4.2) draws on.

Appendix E — Supplementary Figures

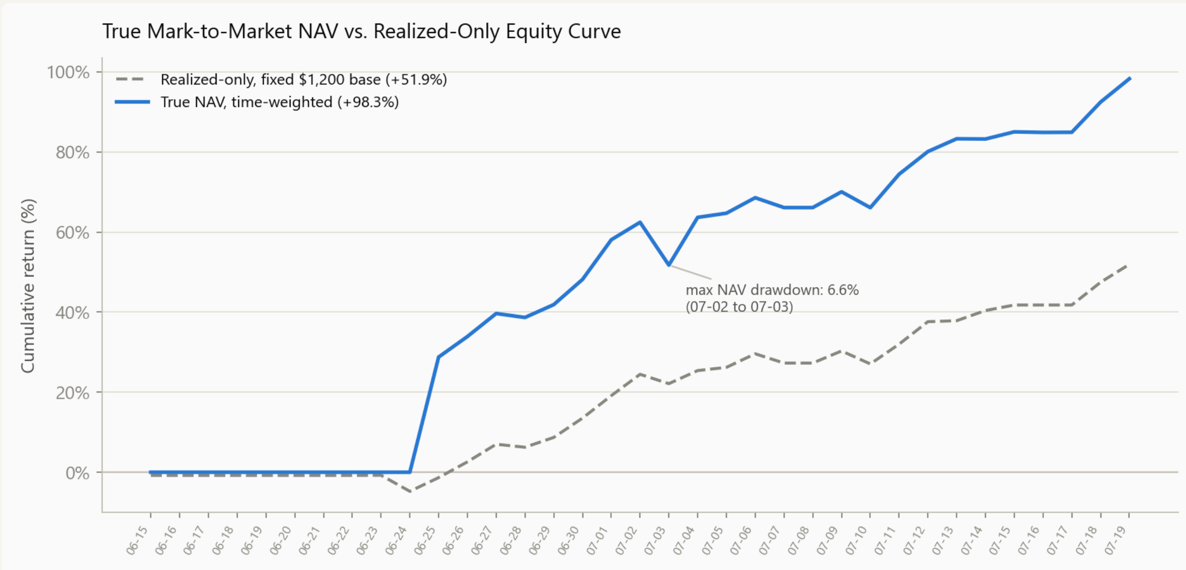
Supporting figures referenced from the main text but not essential to following the paper's core argument on a first read — a redundant results view from §9, and additional risk-metric views from §13. (The microstructure figures that formerly sat here have been moved into §8, where they carry the paper's central claim rather than support it from an appendix.)



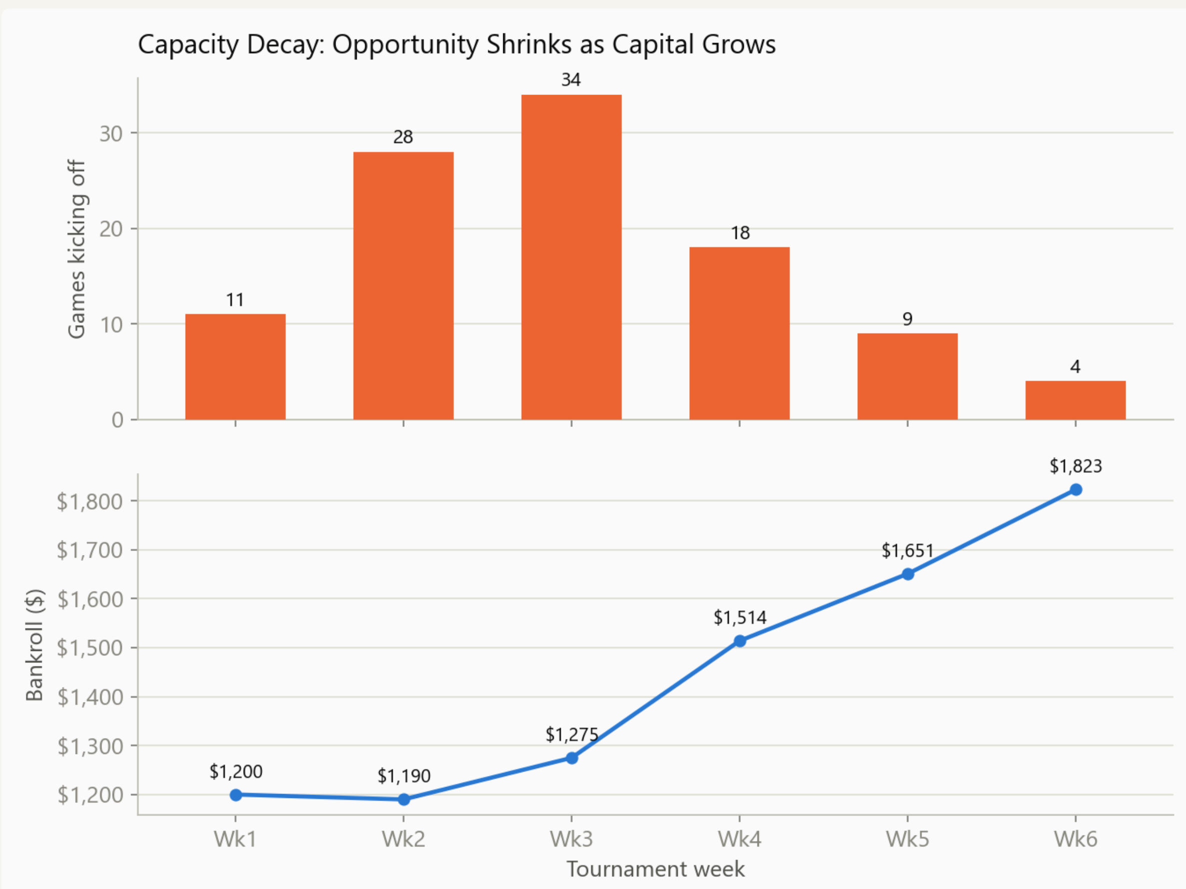
[from §9] Win rate by strategy: I (83.3%), II positive (85.5%), II negative (89.2%).



[from §13] Day-over-day change in the equity-curve series above, colored green on net-positive days and red on net-negative days.



[from §13] Cumulative return, realized-only fixed-base vs. time-weighted true NAV.



[from §13] Weekly game count falling as cumulative bankroll rises — capacity decay in its most direct form.

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